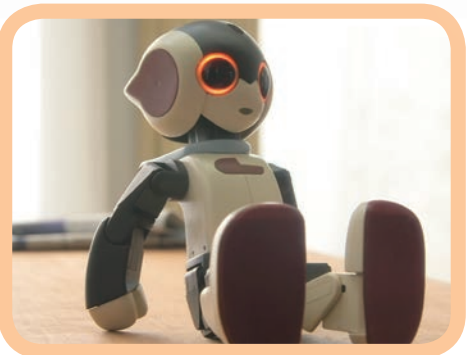


Robi

Build your own

Pack **12**



Stages 43-46:
Build up and test
Robi's limbs

Build your own **Robi**

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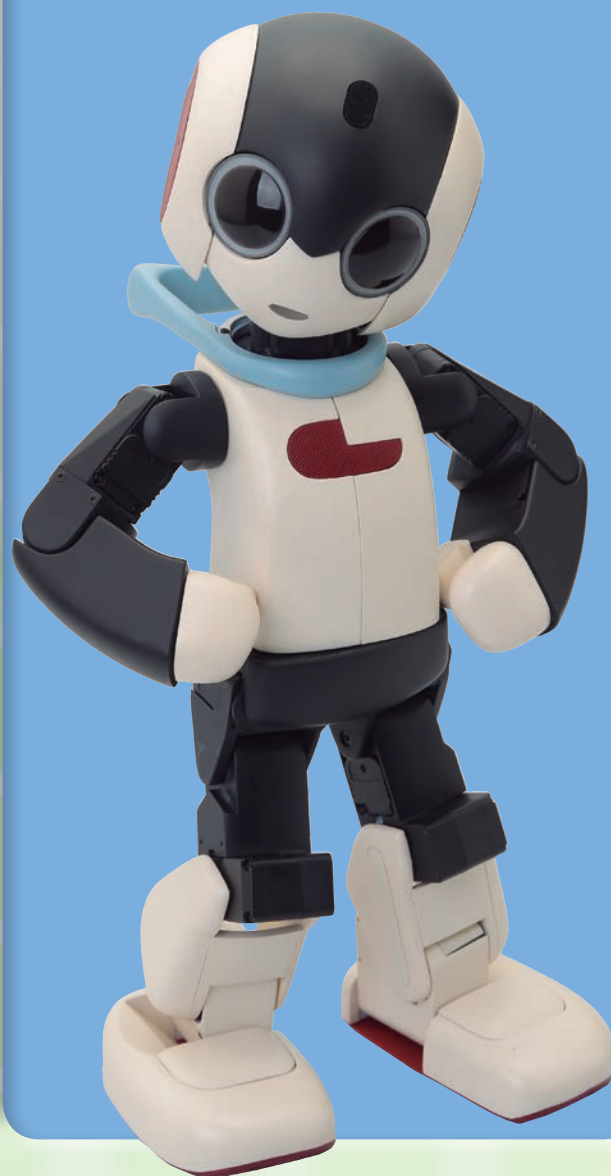
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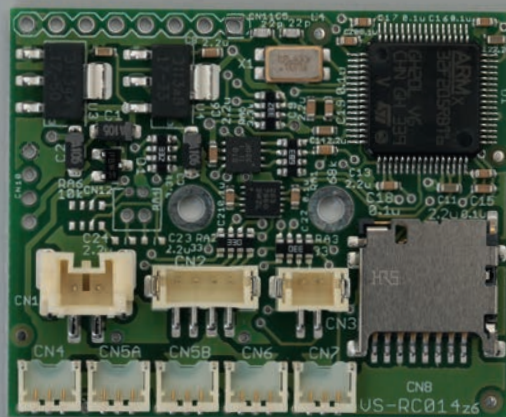


STAGE 43: TEST ROBI'S RIGHT ARM MOVEMENT

Use the test micro SD card supplied with the previous stage and the CPU board supplied with this one to test the movement of Robi's right arm.

The CPU board supplied with this stage is perhaps the most important of Robi's electronic parts. The CPU (Central Processing Unit) is the component that, once fitted with the final micro SD card, will control all of Robi's movements, voice recognition, infrared sensors and the many other features he has. The board houses a number of sockets, into

which will be plugged the cables leading to Robi's various electronic parts, labelled in the diagram on the next page. As well as the sockets, the board has a micro SD card slot, which will be used first to test Robi's movement using the test micro SD card, then, on completion, to hold the final micro SD card.



1

YOUR PARTS

1 CPU board

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

Battery
(Charged in Stage 41)



Right arm assembly
(Stage 14)



Test micro SD card
(Stage 42)



M2 x 4mm pan-head screws x 2 (Stage 41)

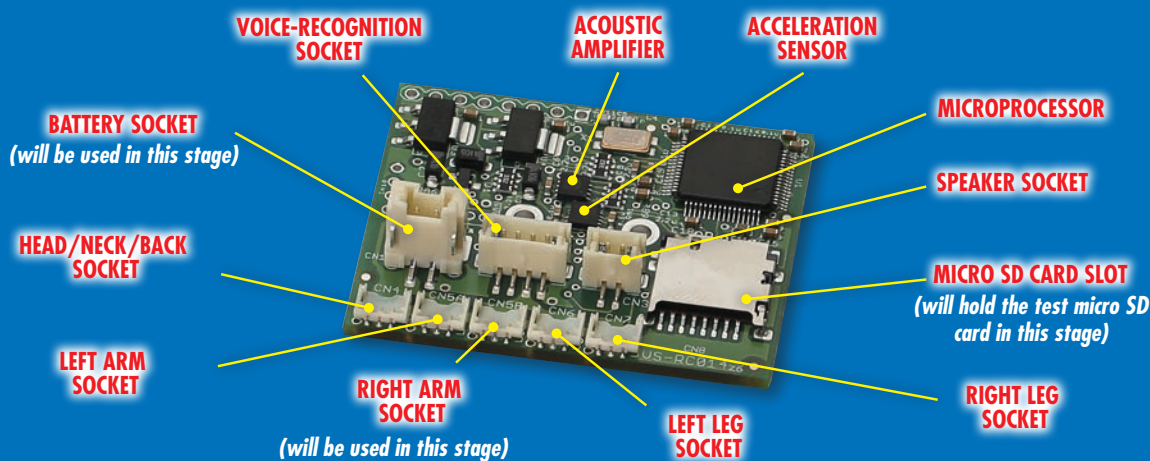
THE CPU BOARD



The best way to understand the role of the Central Processing Unit, or CPU, that you have received in this stage is to see it as Robi's 'brain'. Though the CPU board is built into his body rather than his head, the job it performs is very similar to that of a human brain: it tells Robi's various electronic parts, such as his 20 servos, how and when to act, for example extending his left leg forward while positioning his right one to support his body weight while walking. This complex information, gathered from the micro SD card that fits

into the purpose-built slot on the edge of the board, is decoded by the microprocessor, then carried around his body using a series of cables (much like a human brain uses its nervous system to carry information to the person's limbs).

The CPU board is extremely delicate, and should always be handled with care: avoid touching the electronic components set onto its surface by holding it by its sides; only handle it with completely dry hands; and keep it away from magnetised objects.



Always hold the CPU board by its sides!

WARNING!

When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!

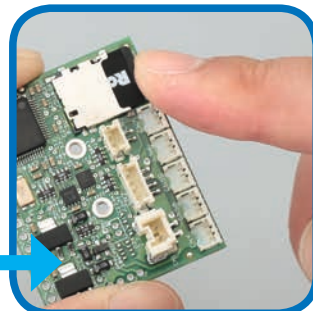
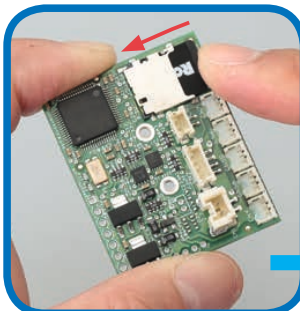


ATTENTION!

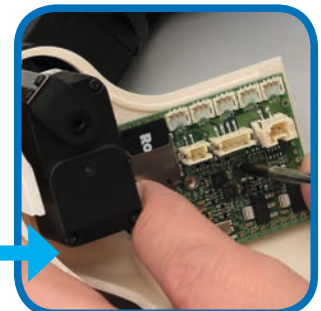
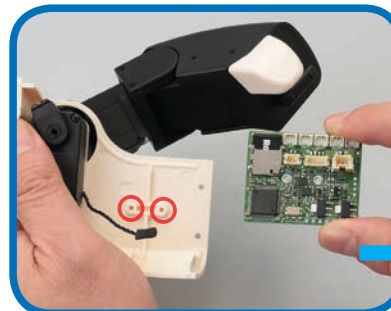
Robi's right arm will start moving as soon as you begin the test, so make sure you do this in a safe place where the arm will not knock anything over or be at risk of damage.

In this stage, you will test the right arm *alone*. Do not attempt to connect any other of Robi's limbs yet: the movement tests must be done as per the instructions!

CONNECTING THE RIGHT ARM

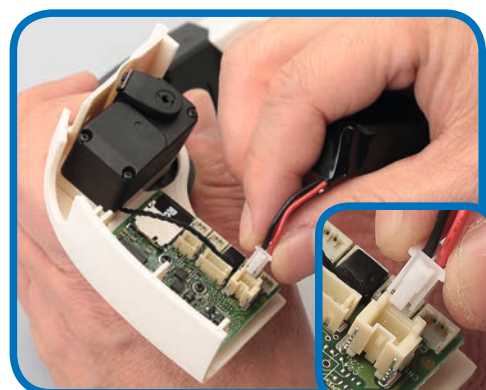
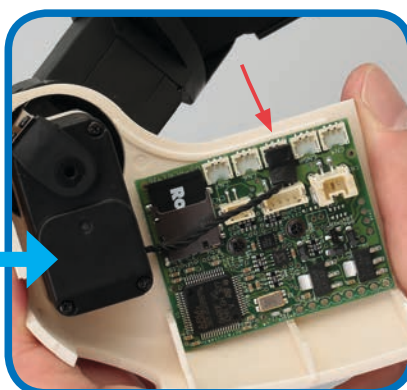
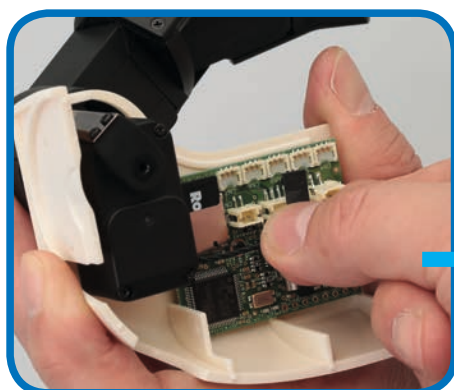


1 Insert the test micro SD card into the card slot on the CPU board, making sure that it clicks fully into place.



2 Line up the CPU board to the two circled holes on the inside of the right body panel. Use the two M2 x 4mm pan-head screws saved from Stage 41 to secure the board to the panel.

THE RIGHT ARM TEST



3 Take the servo cable leading from the right arm's shoulder servo, and press it into the right arm socket on the CPU board. This is arrowed above, and indicated in the diagram on the previous page. As ever, make sure that the connector is set straight and clicks into place neatly and securely, and take care not to force the part.

4 Once the servo cable is connected, plug the battery pack's connector into the battery socket. Again, be very careful not to force the parts. Note: the arm will begin moving as soon as power is supplied.



5 Once the battery is connected, Robi's right arm will immediately begin its test operation. First, the arm will return to its starting position, which resets all the servos to their neutral positions. Then, the forearm servo will start to move, causing Robi's forearm to extend then retract from the elbow, ending up again at the starting position. Next, the upper arm servo will lift Robi's arm outwards before returning it to the starting position. Then, to complete the test, the arm will swing forward and back from the shoulder, again returning to the starting position once done. The test will keep running continuously as long as the battery is connected – give it two or three rounds of the test to ensure that everything is working as it should be, then simply unplug the battery to end the test. [To see a video of this stage's assembly in full, visit the Robi ModelSpace web page at www.model-space.com](http://www.model-space.com)

If the test does not run as described above, read the following pages carefully to see how to address the issue.

TROUBLESHOOTING

As with any device or model that works with a number of electrical components, there is a chance that something might not work as intended the first time it is switched on. If the movement test carried out in this stage does not go as described in Step 5, the most likely cause will be down to an error or a poor connection with one of the right arm's servos. Read the following pages carefully to see how to assess and correct any potential problems.

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:



Servo removal tool (Stage 41)

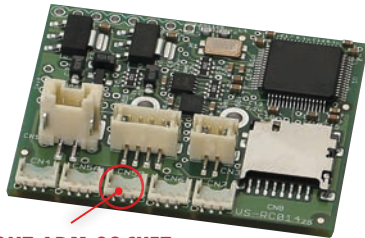


Neck stand and test board (Stage 8)

IF NONE OF THE SERVOS MOVE

Before dismantling the arm to check the individual servos, which can be time-consuming, first check the other electrical components used in the test. Then, if this doesn't solve the problem, continue on to the following pages to see how to dismantle the arm safely and efficiently.

CHECK THAT THE ARM IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



RIGHT ARM SOCKET

This is the middle of the CPU board's five servo sockets, and should accommodate the servo cable leading to the right arm.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure the right arm's servo cable connector is sitting securely within the middle socket, removing and replacing it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure this is fully charged by following the steps explained in Stage 41.

IF ONLY THE FOREARM SERVO MOVES

If the forearm servo moves (bending the arm at the elbow) but either of the other two servos do not, then it is likely that a problem with one or other of these is causing the test to fail.

DISASSEMBLY

No matter which servo is malfunctioning, the arm must be dismantled **from the shoulder**. Also, remember to check each servo's connectors before resetting any ID numbers.

- **The shoulder servo does not move**
Remove the shoulder servo and check/reset its ID number
- **Neither the shoulder nor the upper arm servos move, or the upper arm servo does not move**
Remove the shoulder servo and upper arm servo, and check/reset the ID number(s) and connections as necessary

IF THE FOREARM SERVO OR NO SERVOS MOVE

Whether or not the other arm servos move, if the forearm servo does not, there is most probably an issue with its ID number or cable connections.

DISASSEMBLY

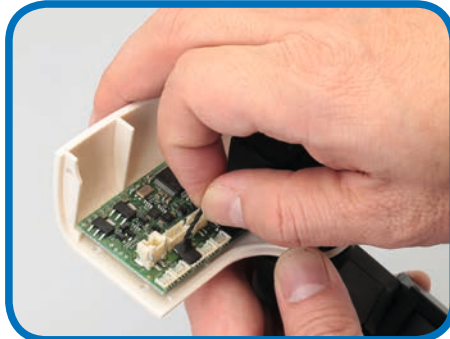
No matter which servo is malfunctioning, the arm must be dismantled **from the shoulder**. Also, remember to check each servo's connectors before resetting any ID numbers.

- **The forearm servo does not move**
Remove the shoulder, upper arm and forearm servo, and check/reset the ID number(s) as necessary, also remembering to check the connections between all servos

DISMANTLING THE RIGHT ARM

The following steps show how to dismantle the right arm in its entirety – but remember that you may only need to remove some of the arm's parts. For instance, if it is the shoulder servo (which gives Robi's arm its forward-and-backward movement and is set on the inside of the right body cover) that needs attention, working **up to Step 9** then testing this servo should be sufficient. If it is the upper arm servo that raises the arm

outwards and upwards that needs attention, work **up to Step 16**. Finally, if the forearm is not working as it should, you will need to work all the way **up to Step 25**. Remember – no matter which servo needs attention, the entire arm must be dismantled, starting at the shoulder. Once you have reset the problem servo(s), refer to the relevant assembly stages, given at the foot of the disassembly pages, to re-build the arm in full.



1 Carefully remove the servo cable from the CPU board.



2 Remove the CPU board from the inside of the right body panel. **Keep the screws safely, making a note of which ones go where.**



3 Remove the shoulder panel, again making sure to keep the screws safely. Once more, make a note of which screws came from where.



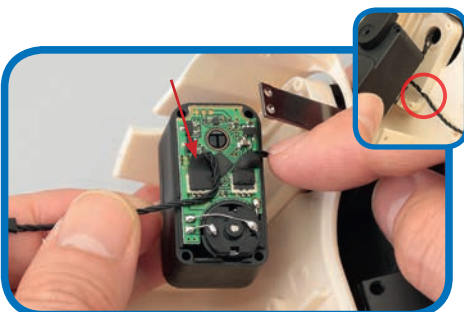
4 Now remove the metal shoulder joint. Keep the screws safely.



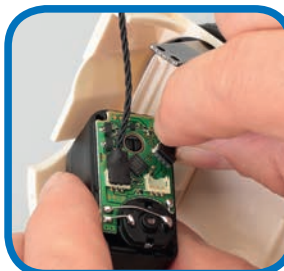
5 Next take out all four screws holding the shoulder servo in place.



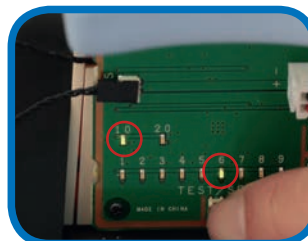
6 Gently lift the servo away from the panel, keeping its cables in place and feeding the exterior cable through the joint to prevent any tugging on the servo's connector socket.



7 Before proceeding with the **disassembly**, check that it is not a connection fault that is causing the problem, and that both servo connectors were in place as they should be. Pay close attention to the cable that bends over on itself (arrowed), as this is more likely to come loose or get caught.



8 If the connectors were both sound, proceed to remove the one leading into the rest of the arm from its socket.



9 Now use the neck stand and test board to check the right shoulder servo's ID number. It should be ID=16. If this is not what you find, refer to the steps in Stage 13 that deal with setting this servo's ID number.

If you are removing more than one servo, make a note of which one each is by using post-it notes with their ID numbers written clearly on them. As well as this, be sure to remember which screws were used for which part of the assembly.

TIP!



10 If necessary, continue dismantling Robi's right arm by lifting the arm away from the body panel entirely.



11 Gently free the metal joint by working the servo cable out of its central hole.



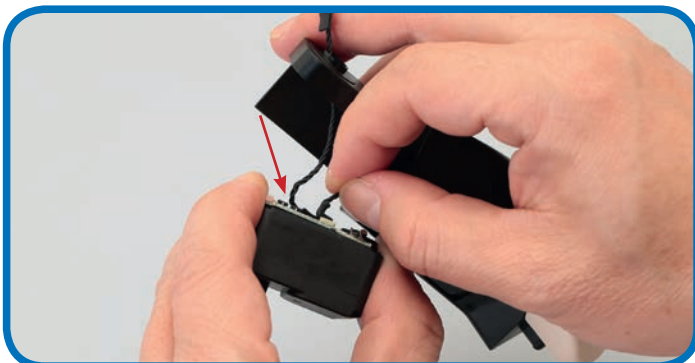
12 Now carefully remove the shoulder bracket from the upper arm servo's shaft.



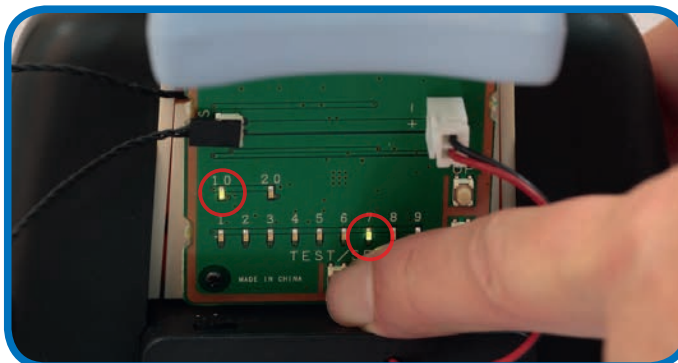
13 Remove the screws from the servo. Keep these safely.



14 Use the servo removal tool supplied with Stage 41 to help you get some leverage on the servo cover, then gently lift this from the servo's body without yanking or tugging on the protruding cable.

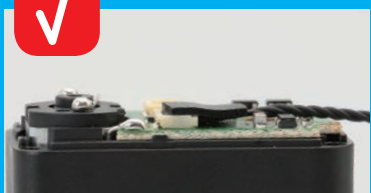


15 Again, check that it is not a basic connection fault that is hindering the test. If the arrowed connector was sound, proceed to remove the other one, leading downwards into the arm and the forearm servo, to free the servo.



16 To check the ID number of the right upper arm servo, connect it to the test board and perform a test as you have done throughout the build. Its number should be ID=17, and if you need to reset this, further information on the resetting process can be found in its original assembly stage, Stage 11.

TIP!



When reconnecting the servo cables' connectors as you perform the tests in this stage, always remember to fit them carefully and properly. They will click into place neatly when aligned correctly – never force them! View the servos from the side to check that the connectors are securely fastened and are flush with the servos' electronic boards.

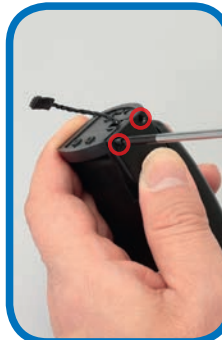
To reassemble the right arm at this point, refer back to Stages 10-12.



17 If you need to look at the forearm servo, proceed by removing the two circled screws holding the upper arm frame in place. Keep the screws safely.



18 Now remove the three screws on the inside of the frame.



19 Unscrew the two screws holding the elbow bracket in place to free the part from the forearm.



20 Remove the three screws holding the forearm cover in place (two on one side and one on the other), then separate the part.



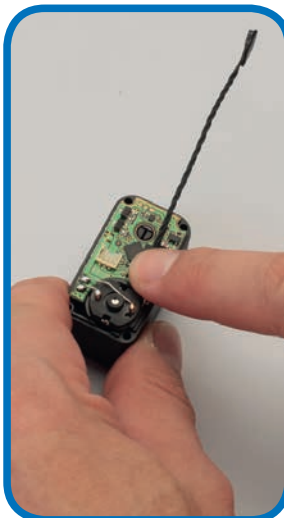
21 Now remove the elbow panel.



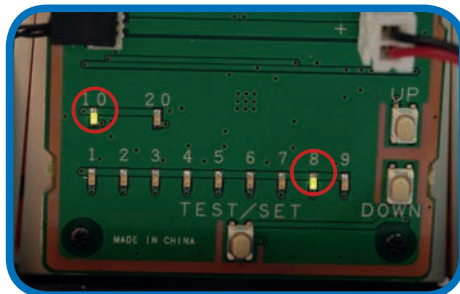
22 Remove the four screws holding the forearm servo in its frame.



23 Again use the servo removal tool to help lift the cover safely from the forearm servo, as shown.



24 Check to see if the servo cable's connector is fitted to its socket as it should be.



25 If the connector was not the problem, test/reset the servo ID number. The right forearm servo should be ID=18. To reset the servo to this, refer back to its original assembly stage, Stage 8.

TIP!



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

To reassemble the right arm at this point, refer back to Stages 8-10.

STAGE 44: PREPARE ROBI'S LEFT ANKLE SERVO AND TEST HIS LEFT ARM

In this assembly, you will prepare, test and set the servo motor that will act as Robi's left ankle, servo ID=10, then use the CPU board to test the movement of the left arm.

This stage sees you returning to the process of preparing, wiring and programming one of Robi's foot servos, this time the left ankle. Once the ID number (10) is set, you will wrap

the servo in white tape to match Robi's exoskeleton, so have this to hand before beginning. You will then repeat last stage's right arm test using the CPU board for the assembled left



1



YOUR PARTS

1 Servo motor (ID=10)

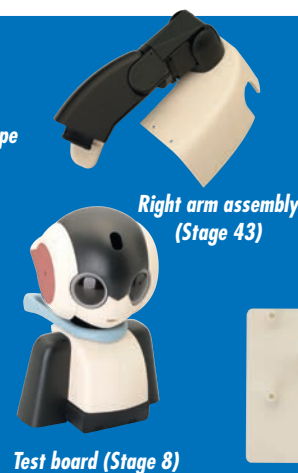
PARTS TO BE ASSEMBLED

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

135mm servo cable
(Stage 37)

White ankle tape
(Stage 25)



Right arm assembly
(Stage 43)



Battery
(Stage 38)



THE SERVO AND CABLE

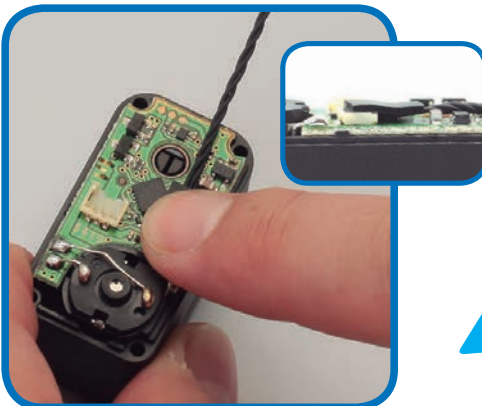
TIP!



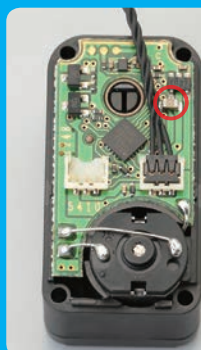
As ever, it is very important that the electronic parts exposed by removing the servo's cover are not touched in any way. Make sure to hold the open servo by its sides, and if you put it down on your work surface during the assembly, do this with the open side facing upwards.



- 1 Take out the four screws holding the servo's removable cover in place, and keep these safely to one side. Then gently lift the cover away, taking care not to force it or damage the circuit board underneath.



TIP!

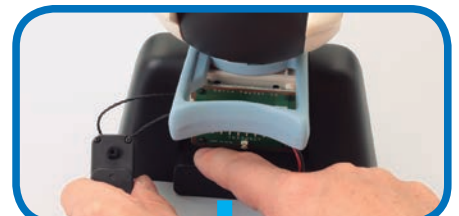
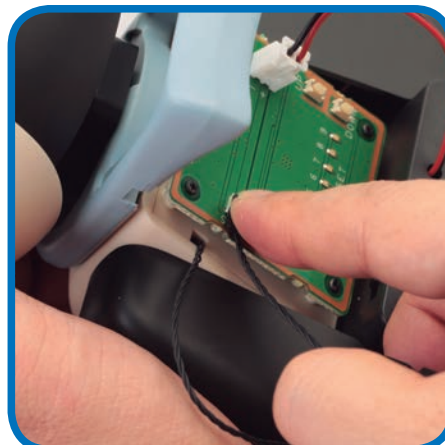


If you find it difficult to attach the cable with the pad in place, it is possible to apply the pad afterwards. If you choose to do this, be especially careful not to press down on the circled projection, next to the cable, as you do so.



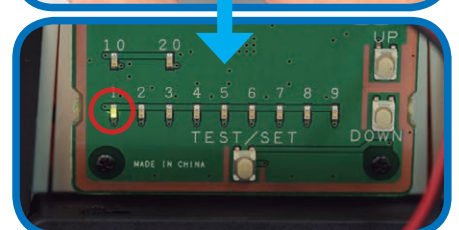
- 2 Fetch the 135mm-long servo cable fitted with protective pads in Stage 37, and press its connector into one of the servo's sockets, as you have done before. Make absolutely sure that the connector is set into place correctly, and that its protective pad is sitting flush to the circuit board, as shown.

- 3 Pass the free end of the servo cable through the circular hole in the removable cover, as shown.

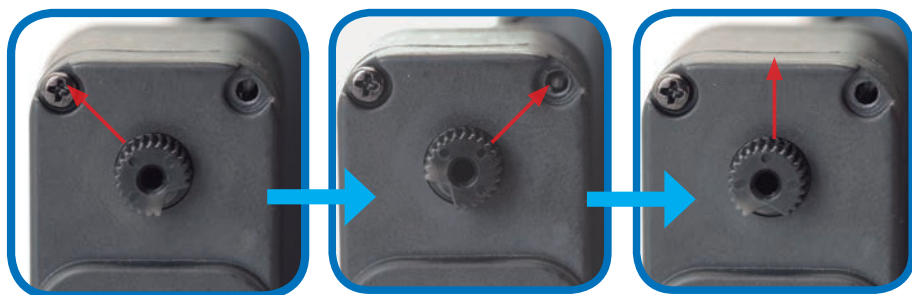


- 4 Replace and secure the cover with all four screws, but only lightly, as you will be removing them again at a later stage.

- 5 Making sure the power is turned OFF on the test board, unplug the existing cable from its socket.



- 6 Attach the free connector of this stage's servo, and turn the power ON. The LEDs should flash twice, then 1 will remain lit.



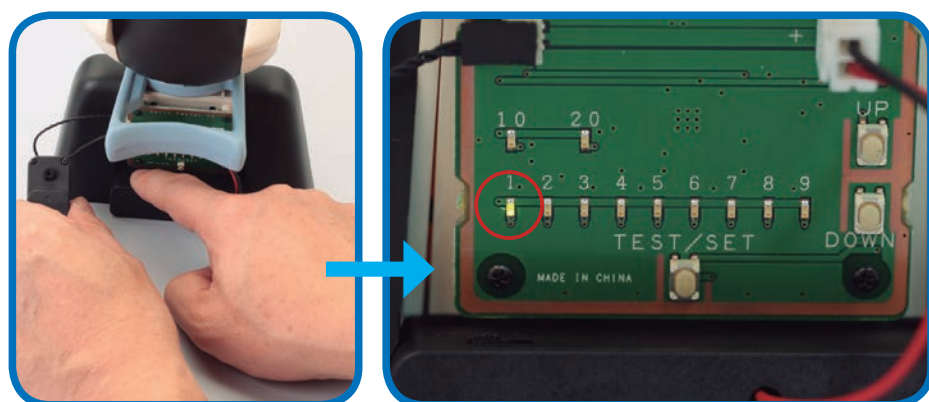
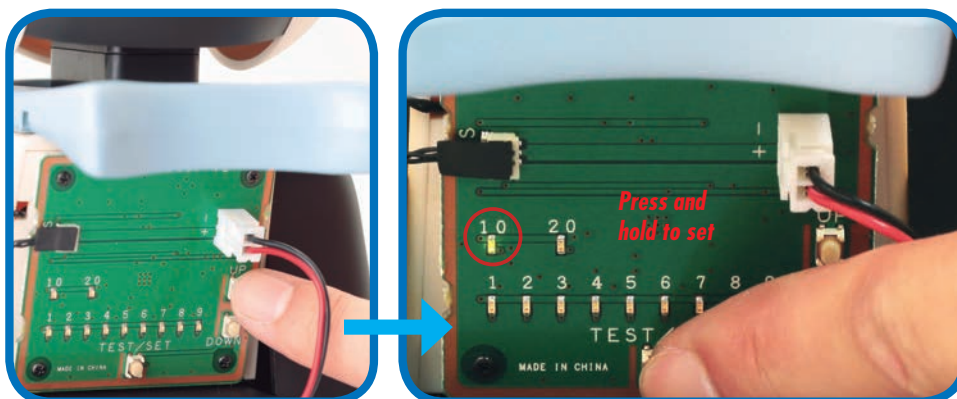
TIP!

If the test does not run when you press the TEST/SET button, or if the LEDs flash rapidly, you may have a faulty connection. Check that the connector is fitted correctly into the test board's socket, and, if it is, carefully unscrew the cover of the servo (replaced in Step 4) and check the connection there. For more info on this, refer to the HELP box in Stage 8.

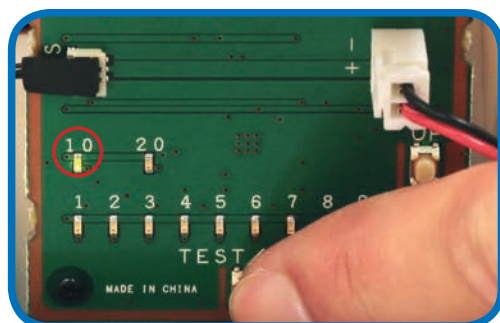
- 7** To test the servo, press the TEST/SET switch: the servo's shaft should rotate 45° to the left, stop, then move back across to 45° to the right. The next step is to set the servo's ID number, so keep the power on.

SETTING THE ID NUMBER

- 8** By now, you will be familiar with the process of setting servos' ID numbers, but refer back to previous stages if you need a reminder. The servo being prepared in this stage will be ID=10, so press the UP button nine times, so that only the 10 LED is lit. When this is the case, press and hold the TEST/SET button for a few seconds. The 10 LED should blink, then remain lit.



- 9** It is vital that Robi's servos are set correctly, so it is always a good idea to check that you have set the intended ID number afterwards. To do this, leave the servo connected, switch the power OFF, then back ON. After all the LEDs have blinked, only the 1 LED should light up.

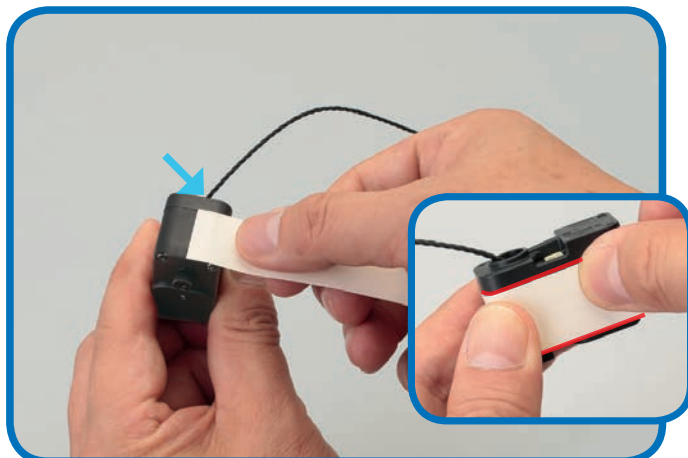


- 10** To check the ID number, press TEST/SET. Now only the 10 LED should light up. Note: if you find that a servo has the wrong number assigned to it, go back and repeat the setting process in full.

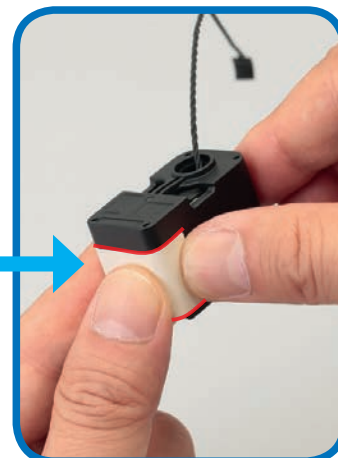
ID=10



APPLYING THE ANKLE TAPE



- 11** Take one strip of ankle tape saved from Stage 25 and apply it, sticky side down, to the side wall of this stage's servo, starting near the shaft. Now start wrapping the tape around the servo's body, following the photo exactly. Make sure no part of the tape extends beyond the edge of the servo (marked by the red lines, inset). It is very important that the tape does not overlap onto the servo's face in any way, as this will inhibit the operation of the moving parts that will attach there.



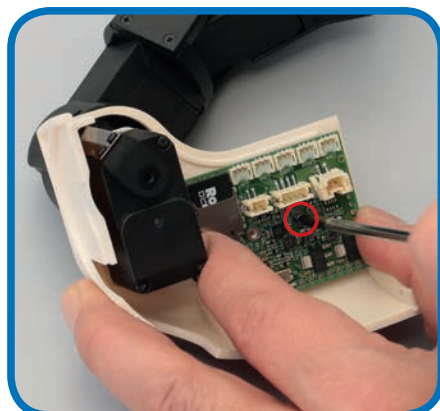
- 12** Still making sure no tape extends beyond the side of the servo, keep wrapping the tape around until it is finished.



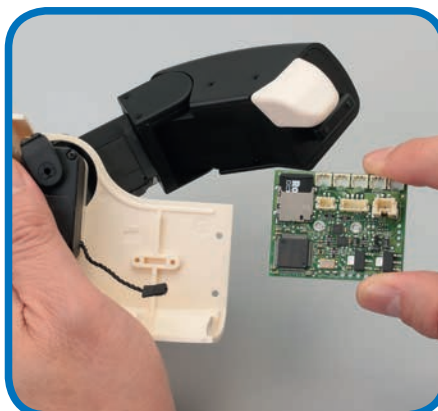
Prepared ankle servo

Robi's left ankle servo is now set with its ID number (ID=10), it is wired, and has its tape applied, ready to be fitted in coming stages.

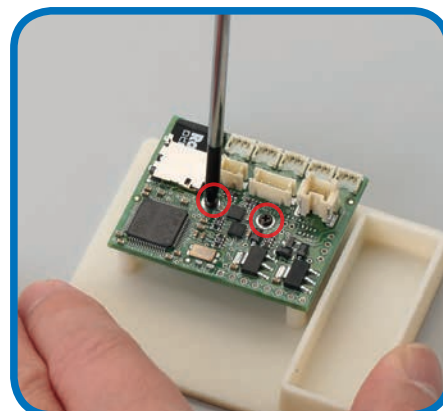
TESTING THE LEFT ARM



- 13** Take the right arm and body panel and remove the two screws holding the CPU board in place.



- 14** Lift the board away from the panel, holding it by its sides so as not to risk damaging the electronic components on its face.



- 15** Use the same two screws to secure the CPU board to its support, saved from Stage 41. **Read the advice below before proceeding.**



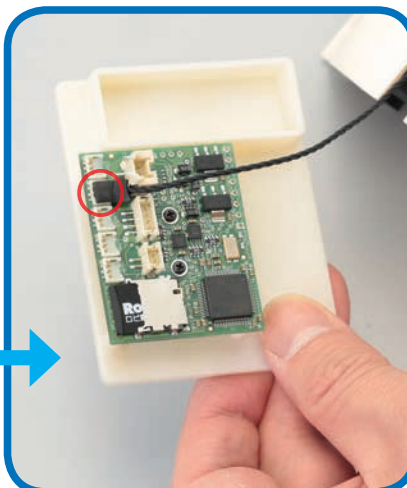
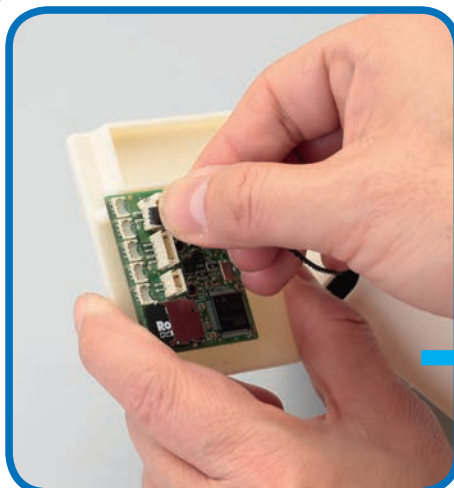
WARNING!

When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!

ATTENTION!

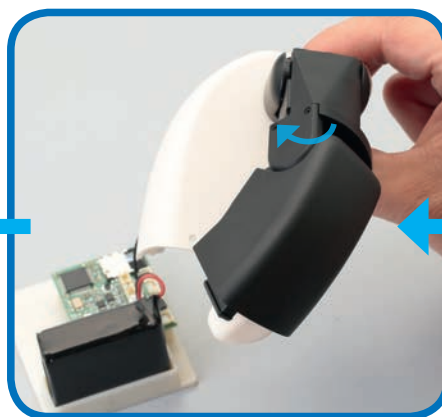
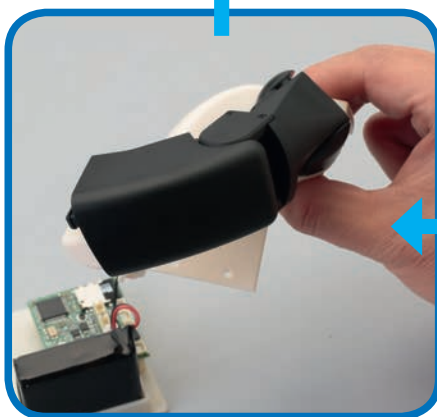
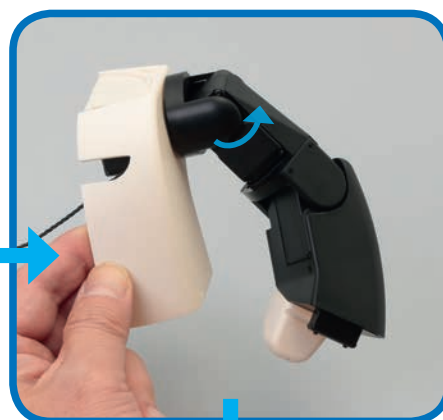
Robi's left arm will start moving as soon as you begin the test, so make sure you perform this test in a safe place where it will not knock anything over.

In this stage you will test the left arm *alone*. Do not attempt to connect any other of Robi's limbs; the movement tests must be done as per the instructions!



16 Take the servo cable leading to the left arm and fit it securely into the circled socket, the second furthest away from the micro SD card slot on the board (Stage 43 includes a diagram listing which socket holds which servo cable). Make sure it is properly connected.

17 Seat the battery in its holder, then connect it to the CPU. Note: the arm will begin moving as soon as power is supplied.



18 Once the battery is connected, Robi's left arm will immediately begin its test operation. First, the arm will return to its starting position, which resets all the servos to their neutral positions. Then, the forearm servo will start to move, causing Robi's forearm to extend then retract, ending up again at the starting position. Next, the upper arm servo will lift Robi's arm outwards before returning it to the starting position. Then, to complete the test, the arm will swing forward and back from the shoulder, again returning to the starting position once done. The test will keep running continuously as long as the battery is connected – give it two or three rounds of the test to ensure that everything is working as it should be, then simply unplug the battery to end the test. [To see a video of this stage's assembly in full, visit the Robi ModelSpace web page at www.model-space.com](http://www.model-space.com)

If the test does not run as described, read the troubleshooting section in this stage to see how to address the issue.

TROUBLESHOOTING

As you may have found with the right arm, there is a chance that something does not work as intended the first time you test the left one. If the movement test carried out in this stage does not go as described in Step 18, the most likely cause will be an error with one of the arm's servos. Read the following pages carefully to see how to assess and correct any potential issues.

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:



Servo removal tool (Stage 41)

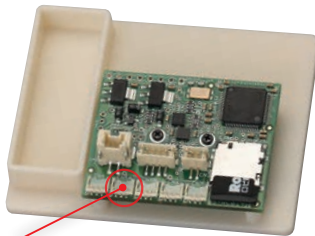


Neck stand and test board (Stage 8)

IF NONE OF THE SERVOS MOVE

Before dismantling the arm to check the individual servos, which can be time-consuming, first check the other electrical components used in the test. If that does not solve the problem, continue on to the following pages to see how to dismantle the arm safely and efficiently.

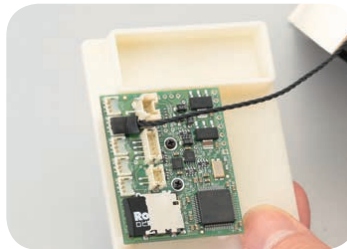
CHECK THAT THE ARM IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



LEFT ARM SOCKET

The socket second-furthest from the CPU's micro SD card slot accommodates the servo cable leading to the left arm.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the left arm's servo cable connector is sitting securely within its socket, removing and replacing it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged by following the steps explained in Stage 41.

IF ONLY THE FOREARM SERVO MOVES

If the forearm servo moves, bending the arm at the elbow, but either of the other two do not, then a problem with one or other of these is probably causing the test to fail.

DISASSEMBLY

No matter which servo is malfunctioning, the arm must be dismantled **from the shoulder**. Also, remember to check each servo's connectors before resetting any ID numbers.

- The shoulder servo does not move
Remove the shoulder servo and check/reset its ID number
- Neither the shoulder nor the upper arm servos move, or the upper arm servo does not move
Remove the shoulder servo and upper arm servo, and check/reset the ID number(s) and connections as necessary

IF THE FOREARM SERVO OR NO SERVOS MOVE

Whether or not the other arm servos move, if the forearm servo does not, there is most probably a problem with its ID number or cable connections.

DISASSEMBLY

No matter which servo is malfunctioning, the arm must be dismantled **from the shoulder**. Also, remember to check each servo's connectors before resetting any ID numbers.

- The forearm servo does not move
Remove the shoulder, upper arm and forearm servo, and check/reset the ID number(s) as necessary. Also remember to check the connections between all servos

DISMANTLING THE LEFT ARM

The following steps show how to dismantle the left arm in its entirety – but remember that you may only need to remove some of the arm's parts. For instance, if it is the shoulder servo (which gives Robi's arm its forward-and-backward movement and is set on the inside of the left body cover) that needs attention, working **up to Step 7** then testing this servo should be sufficient. If it is the upper arm servo, which raises the arm

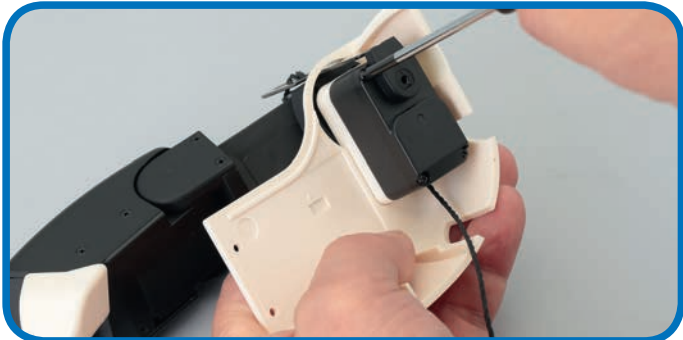
outwards and upwards, that needs attention, work **up to Step 14**. Finally, if the forearm is not working as it should, you will need to work all the way **up to Step 23**. Remember – no matter which servo needs attention, the arm must first be dismantled starting at the shoulder. Once you have reset the problem servo(s), refer to the indicated assembly stages to reassemble Robi's left arm in full.



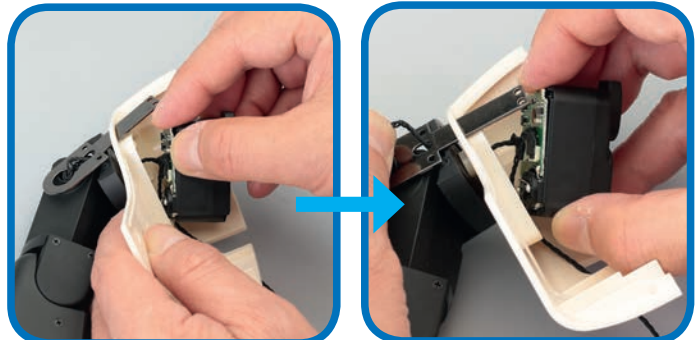
- 1** Carefully remove the shoulder panel by taking out the two screws holding it in place. **Keep these screws, remembering where they came from.**



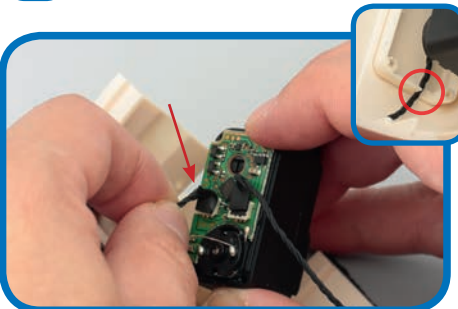
- 2** Now remove the metal shoulder joint in the same way, again remembering to keep the screws safely.



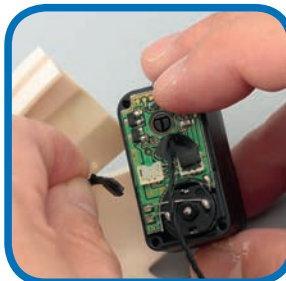
- 3** Now remove the four screws holding the servo in place. Keep the screws.



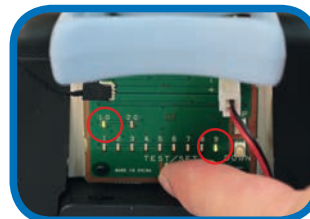
- 4** Lift the servo away very carefully, feeding the exterior cable through the hole in the joint to ensure that it doesn't get caught or tug on the servo's connector socket.



- 5** Before proceeding, check that it is not a connection fault and that the connectors of both cables were in place as they should be. Pay close attention to the cable that bends over on itself (arrowed), and check that the groove in the frame is not trapping its cable (inset).



- 6** If the connectors were both in place correctly, proceed to remove the one leading into the rest of the arm from its socket.



- 7** Now use the test board to check the left shoulder servo's ID number. It should be ID=19. If this is not what you find, refer to the steps in Stage 21 that deal with setting this servo's ID number.

If you are removing more than one servo, make a note of which one each is by using post-it notes with their ID numbers written clearly on them. As well as this, be sure to remember which screws were used for which part of the assembly.

TIP!

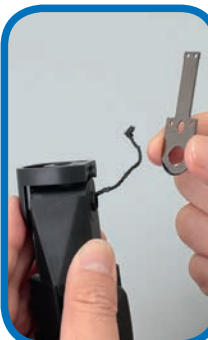
To reassemble the left arm at this point, refer back to Stages 20-22.



8 If necessary, continue dismantling Robi's left arm by lifting the arm away from the body panel entirely.



9 Gently work the metal joint free of the servo cable by first pulling this out of its central hole.



10 Now remove the shoulder bracket from the upper arm servo's shaft.



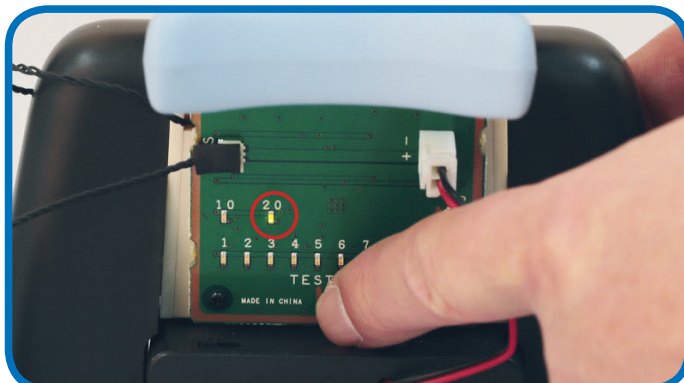
11 Remove the screws from the servo. Keep these safely.



12 Use the servo removal tool supplied with Stage 41 to help remove the servo's cover gently without yanking or tugging on the protruding cable.



13 First check that it is not a basic connection fault that is hindering the test. If the arrowed connector was sound, proceed to remove the other one, leading downwards into the arm and the forearm servo, to free the servo.



14 To check the ID number of the left upper arm servo is correct, connect it to the test board and perform a test as you have done previously. If you need to reset it, its number should be ID=20, and further information on resetting this can be found in its original assembly stage, Stage 19.

TIP!



When reconnecting the servo cables' connectors as you perform the tests in this stage, always remember to fit them carefully and properly. They will click into place neatly when aligned correctly – never force them! View the servos from the side to check that the connectors are securely fastened and are flush with the servos' electronic boards.

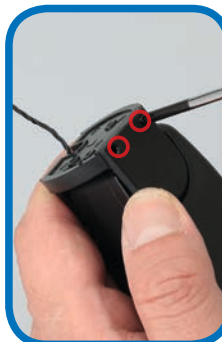
To reassemble the left arm at this point, refer back to Stages 19-20.



- 15** If you need to look at the forearm servo, proceed by removing the two circled screws holding the upper arm frame in place. **Keep the screws.**



- 16** Now remove the three screws on the inside of the frame.



- 17** Unscrew the two screws holding the elbow bracket in place to free it from the forearm.



- 18** Remove the three screws holding the forearm cover in place (two on one side and one on the other), then remove the cover.



- 19** Now remove the elbow panel.



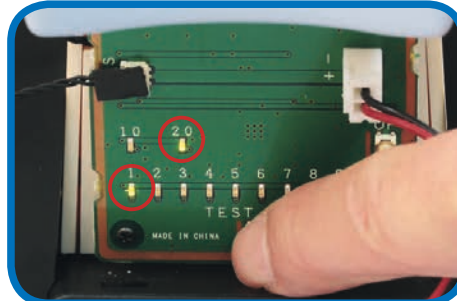
- 20** Remove and set aside the four screws holding the forearm servo in its frame.



- 21** Use the servo removal tool to help lift the cover safely from the forearm servo, as shown.



- 22** Check to see if the servo cable's connector is fitted to its socket as it should be.



- 23** If the connector was not the problem, test/reset the servo ID number. The left forearm servo should be ID=21. To reset the servo to this, refer back to its original assembly stage, Stage 16.

TIP!



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

To reassemble the left arm at this point, refer back to Stages 16-18.

STAGE 45: BUILD UP ROBI'S LEFT FOOT AND TEST ITS SERVOS

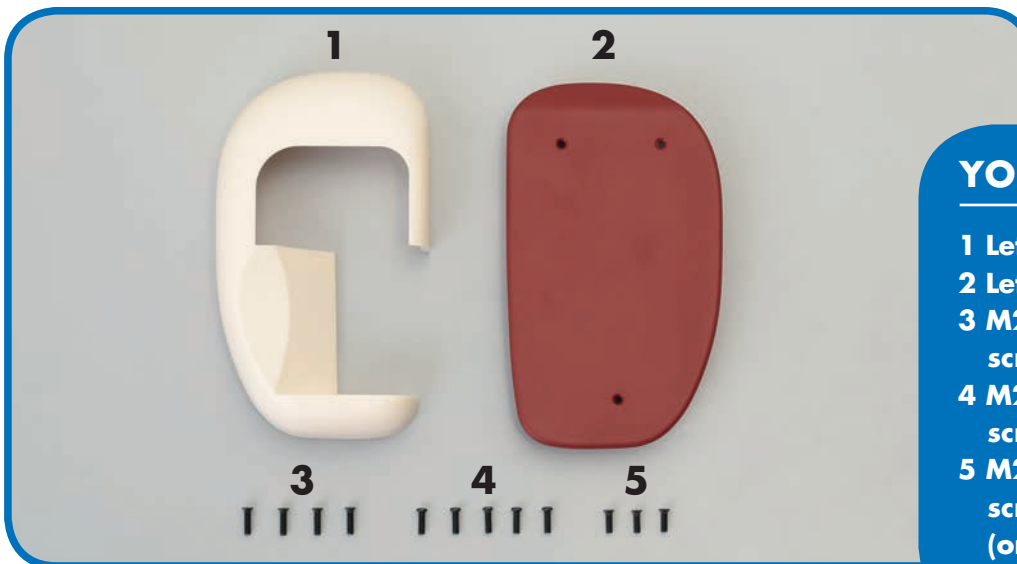
In this session, you will build up Robi's left foot by adding the ankle servo, prepared over the previous stages, to the rest of the foot, before testing its movement with the CPU board.

This stage sees you connect Robi's left foot and ankle servos in much the same way as you did for the right foot back in Pack 8. However, this time you will also be linking the assembly up to the CPU board to test the parts' movement at the end of this stage.

As ever, take great care when handling exposed electronic components, and follow the steps closely – even if they seem identical to those in past stages. A troubleshooting section is found at the end of the stage to help you correct any errors you may encounter.



PARTS TO BE ASSEMBLED

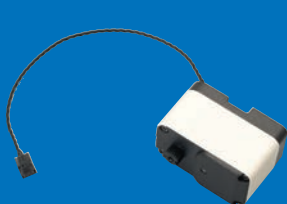


YOUR PARTS

- 1 Left foot side cover
- 2 Left sole
- 3 M2 x 7mm countersunk screws x 4
- 4 M2 x 6mm pan-head screws x 5
- 5 M2 x 6mm countersunk screws x 3
(one of each type of screw is a spare)

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.



Taped and wired ankle servo (Stage 44)



Foot servo and ankle frame (Stage 37)



Left ankle side panel and joint (Stage 36)

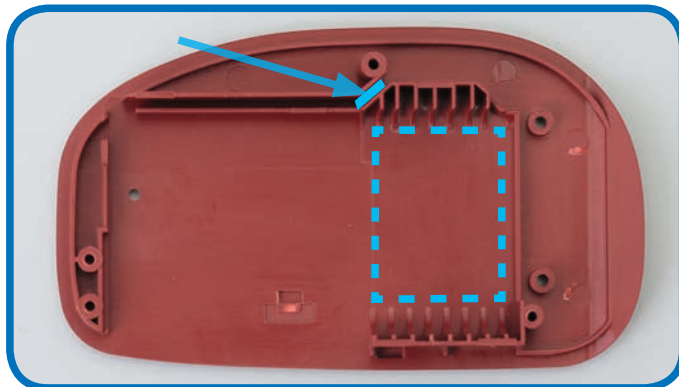


Battery (Stage 38)

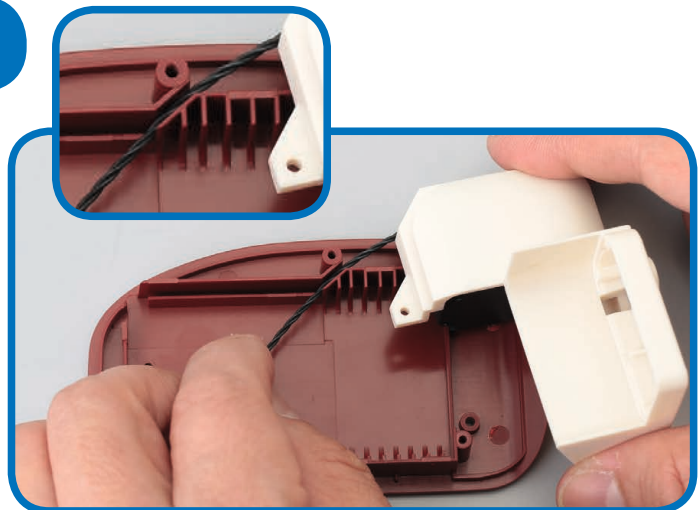


CPU board and support (Stage 44)

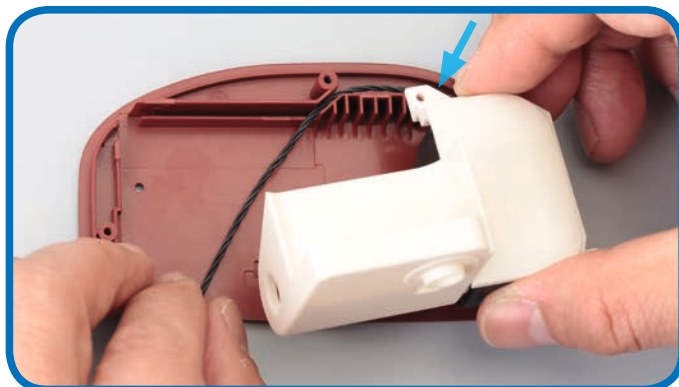
FITTING THE FOOT SERVO



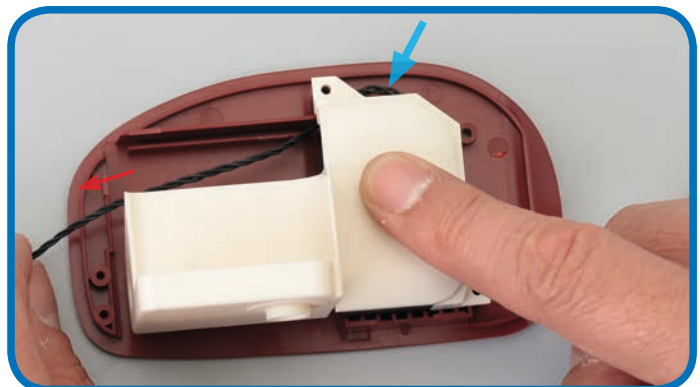
- 1 Familiarise yourself with the inside of the left sole. The first servo you will fit in this stage will be placed in the area marked with dotted blue lines. The angled groove (arrowed) will accommodate the servo's cable once it is fitted.



- 2 Take the foot servo and ankle frame (Stage 37) in one hand, and with the other, fit the servo cable into the angled groove highlighted in the previous step.



- 3 Making sure that the cable doesn't come out of the groove, lower the servo, positioned as shown, into the rectangular space highlighted in Step 1. Pull the cable gently to the left as you do this, to minimise slack.



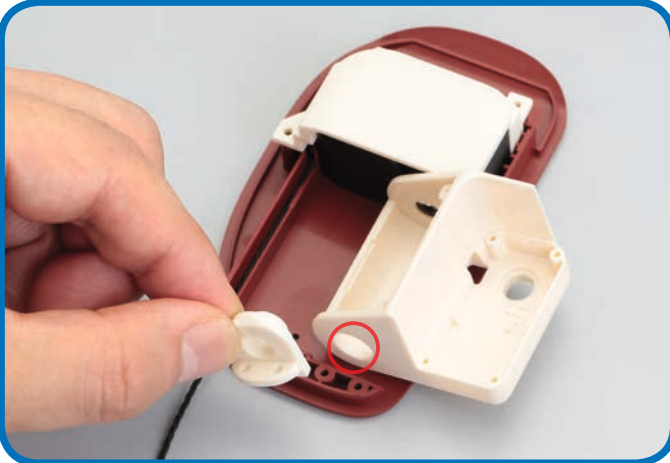
- 4 Press the servo into the space so that it fits snugly, pulling the servo cable gently to the left to ensure that no slack forms and that the cable fits easily within the contour of the sole.



- 5 Now you need to eliminate the slack at the servo end completely, which is done by gently pulling the free end of the cable through the groove (hold the cable itself – do not pull on the connector). Be very careful not to tug on the cable too hard, as this could damage it or cause it to come free inside the servo, which will mean opening the servo and resecuring the cable. If the cable comes out of the groove as you pull it, simply push it back in.



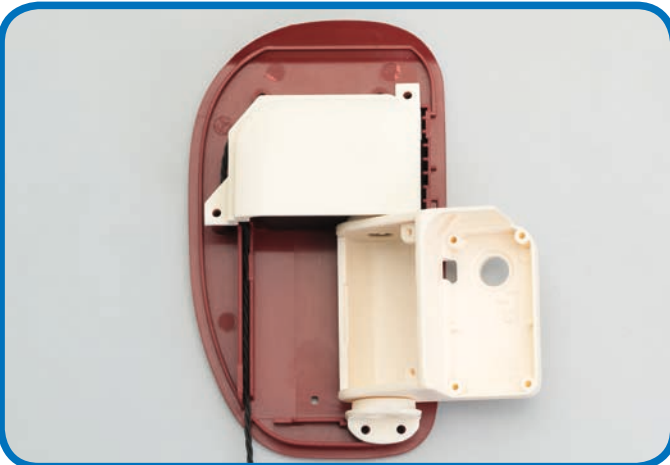
- 6 Hold the foot servo and its cover in place, then tilt the ankle frame over to expose its inner side, as shown.



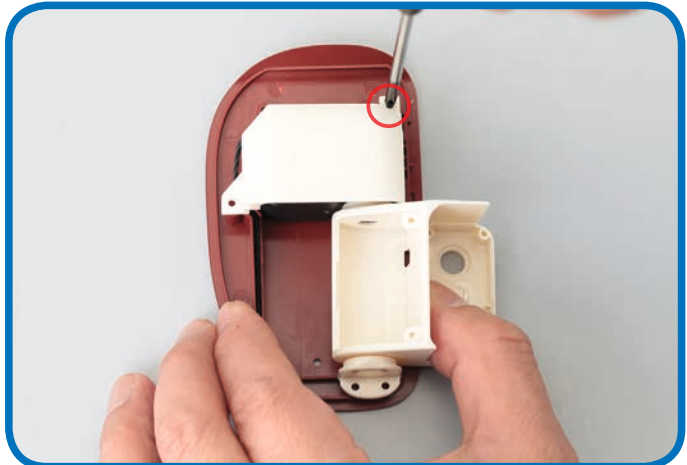
- 7** Take the left ankle joint, saved from Stage 36, and line it up to the circled hole at the rear face of the opened ankle frame. Align the part as shown.



- 8** Hold the ankle frame in place with one hand, and, with the other, push the joint into the hole so that your assembly looks like the one in the photo.



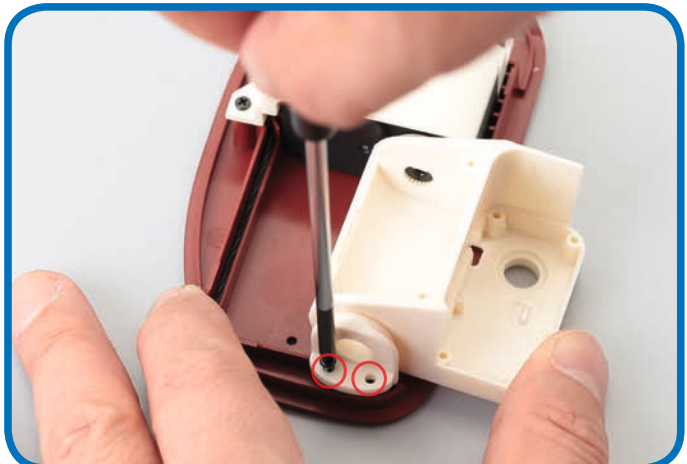
- 9** Your assembly should look like this when seen from above. Check that the cable is still set neatly within the groove, and that the joint fitted in Step 7 is sitting exactly as shown.



- 10** Hold the ankle frame in position again, then tighten an M2 x 6mm pan-head screw into the circled hole.

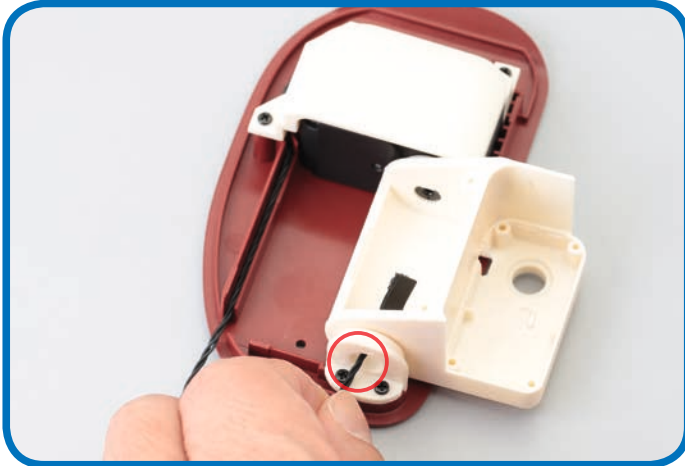


- 11** Now tighten another M2.6 x 6mm pan-head screw into the circled hole on the other side of the cover.



- 12** Use the remaining two M2 x 6mm pan-head screws to secure the ankle joint to the sole, as shown.

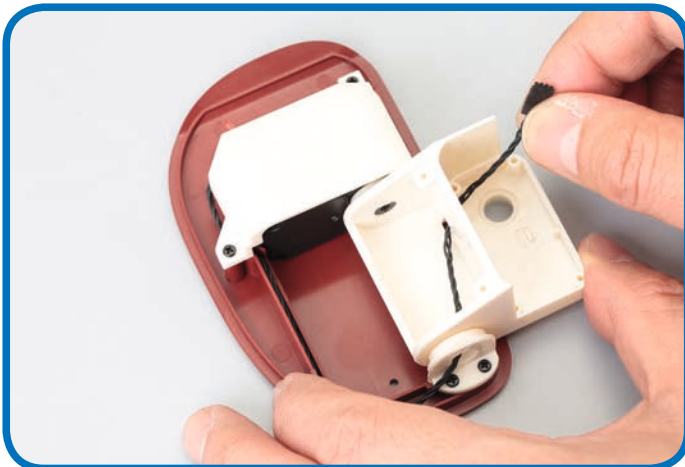
FITTING THE ANKLE SERVO



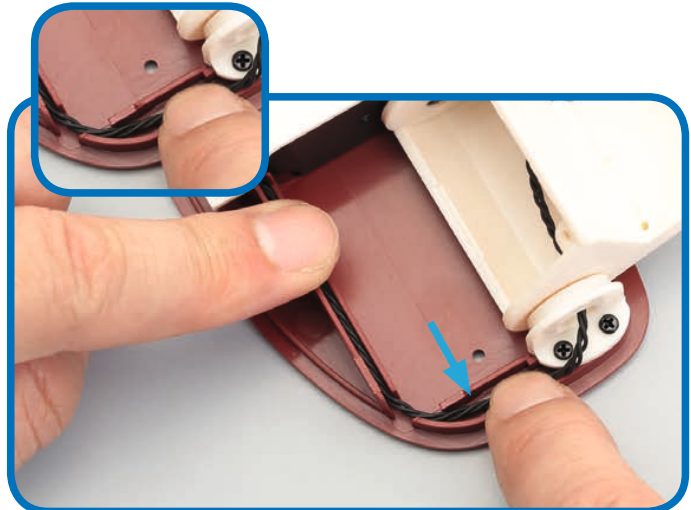
- 13** Feed the foot servo's cable around and through the hole in the ankle joint.



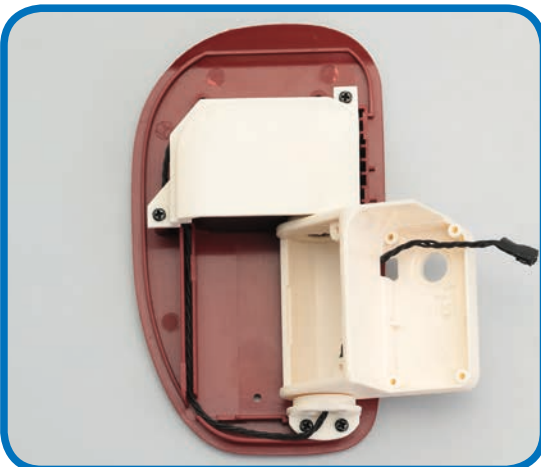
- 14** Then ease it through the rectangular hole in the ankle frame, working from the inside.



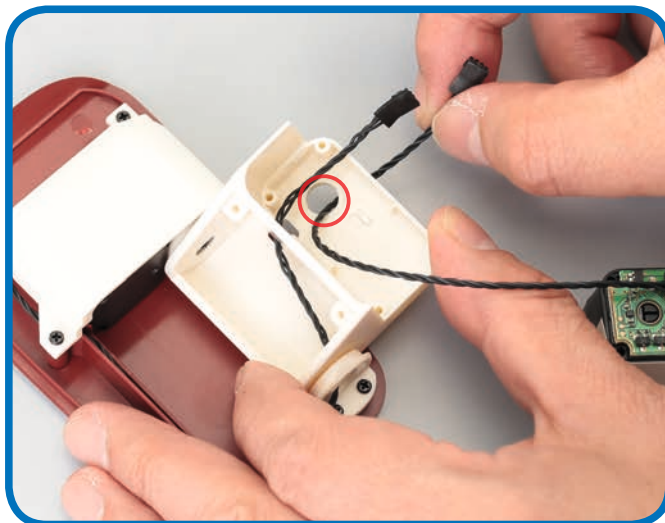
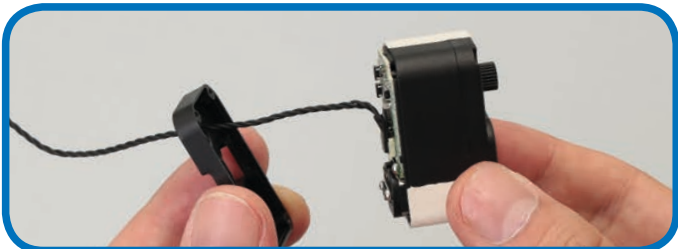
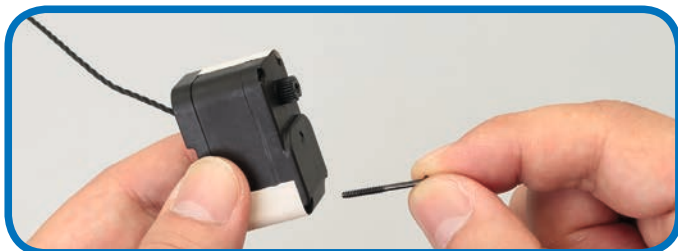
- 15** Gently pull the cable through to eliminate any slack.



- 16** Fit the servo cable into the indicated groove between the sole's rectangular flat and the exterior wall, so that the space is clear.

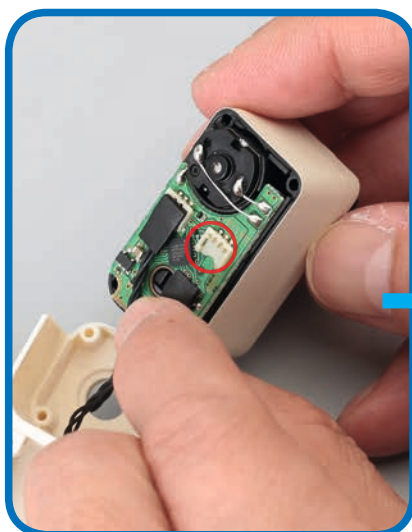


- 17** Your assembly should look like this, with the cable only leaving the groove to rise into the ankle joint and frame.



18 Remove the cover of the taped and programmed ankle servo from Stage 44, by unscrewing the four screws holding it in place. Keep these safely.

19 Feed the ankle servo's cable through the circular hole in the ankle frame from the inside of the part.

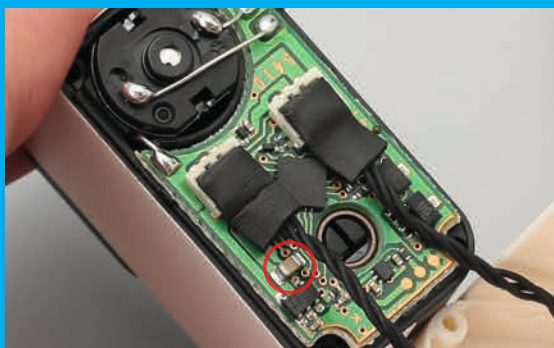


20 Now take the servo cable leading to the foot servo, fitted to the sole in Step 14, and press its connector into the free socket on the ankle servo, circled in the photo far left. Double check that both connectors are set securely.

TIP!

If you find it difficult to attach the cables with their protective pads already in place, it is possible to apply these afterwards. If you choose to do this, be especially careful not to press down on the circled projection, next to the cable, as you do so.

When reconnecting the servo cables' connectors as you perform the tests in this stage, always remember to fit them carefully and properly. They will click into place neatly when aligned correctly – never force them! View the servos from the side to check that the connectors are securely fastened and are flush with the servos' electronic boards.



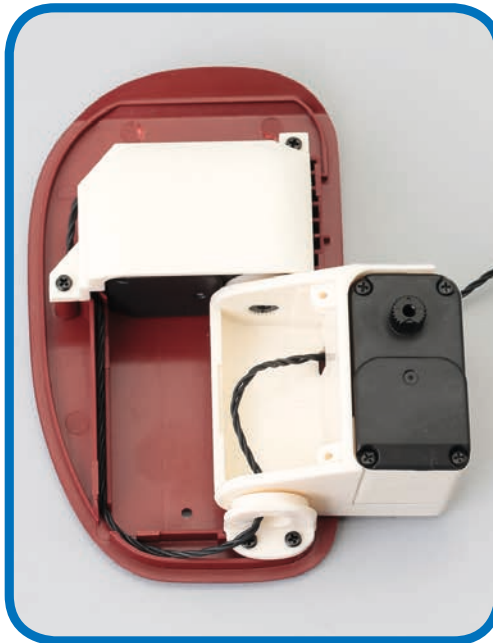


- 21** Lower the ankle servo, aligned as shown, into the ankle frame, feeding the servo cables through the circular hole (red arrow, above left) and through the rectangular hole (blue arrow, above right) so that there is no wire bunched up inside the servo frame.



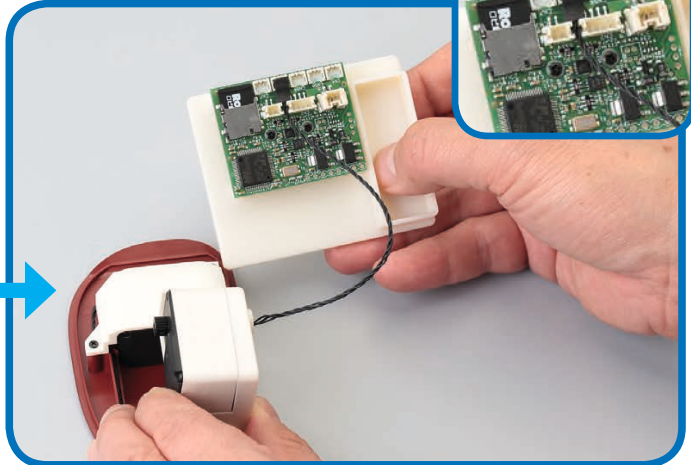
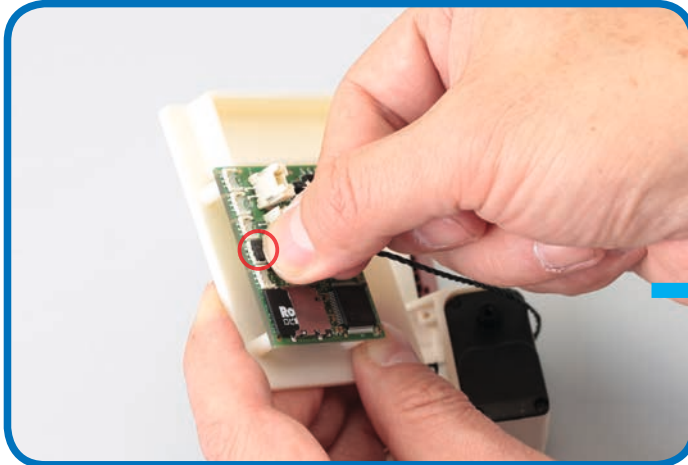
- 22** Press the servo into the frame so that it fits snugly. There should be no gaps visible between the edge of the ankle frame and the white tape wrapped around the servo.

- 23** Double-check that there is no slack inside the servo by pulling the indicated cable through again before proceeding to fasten the servo in place.



- 24** Turn the frame so that the servo faces upwards, then use the four screws removed in Step 18 to secure it to the ankle frame, as shown.

TESTING THE ANKLE AND FOOT SERVOS



- 25** Fetch the CPU board, last used in Stage 44, and connect the servo cable leading to the left foot and ankle assembly built in this stage to the circled socket, second closest to the micro SD card slot. You'll find a diagram showing where each connector goes in Stage 43.



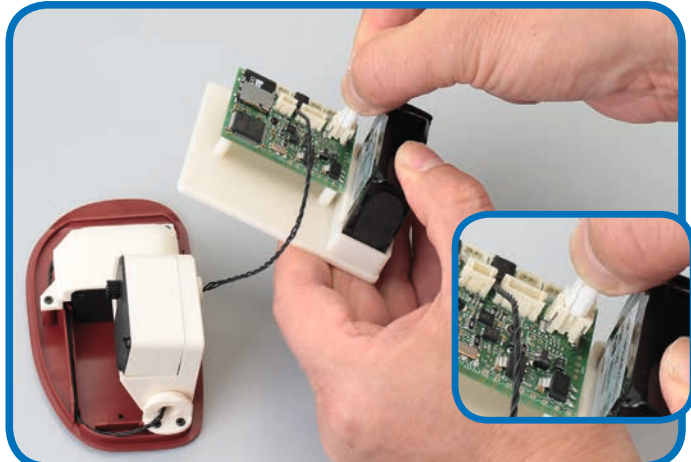
WARNING!

When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!

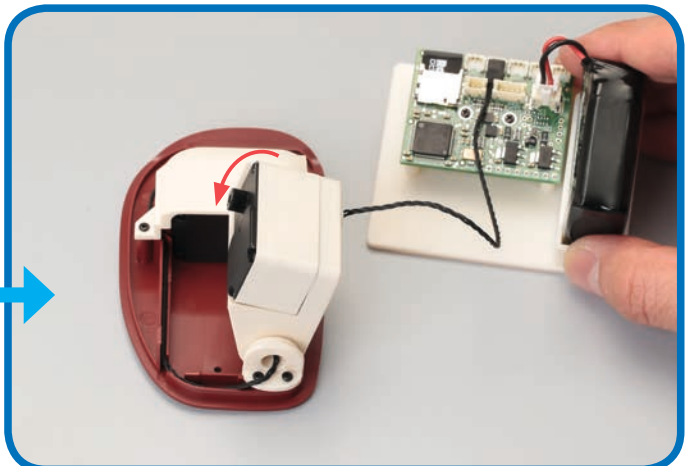
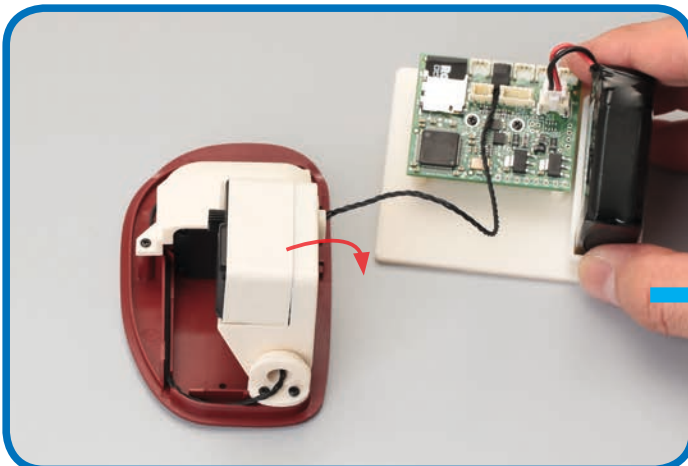
ATTENTION!

Robi's left foot will start moving as soon as you begin the test, so make sure you perform this test in a safe place where it will not knock anything over.

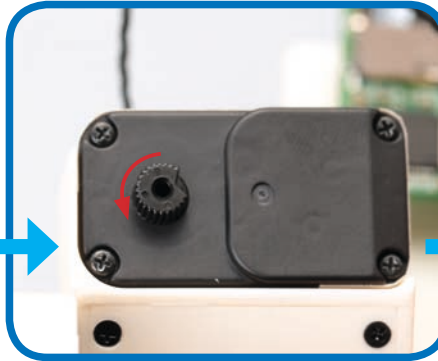
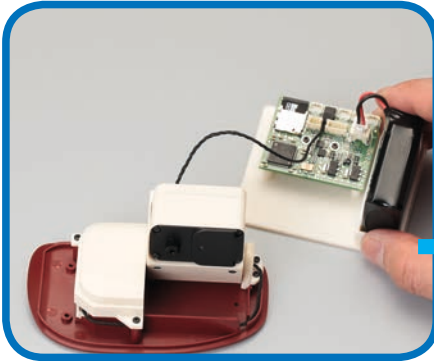
In this stage, you are testing the left foot *alone*. Don't try to connect any other of Robi's limbs; the movement tests must be done as per the instructions!



- 26** Place the left foot on a flat surface with adequate space around it. Then connect the battery to its designated socket (inset). Remember, the foot will begin moving as soon as power is connected, so be prepared.



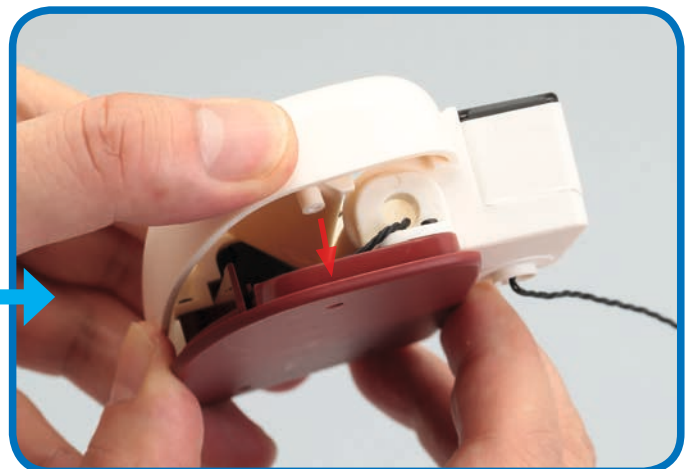
- 27** To begin the test, the foot servo will move to its neutral position, causing the ankle frame attached to it to straighten up to be set perpendicular to the sole. Then, it will bend over from side to side once.



- 28** When the ankle frame has returned to its neutral position, the shaft of the ankle servo that it holds will rotate. You will have to look closely to see this, as the shaft is not yet attached to a larger part or frame. The test will run continuously until you remove the battery from the CPU board, so simply do this to stop the process.

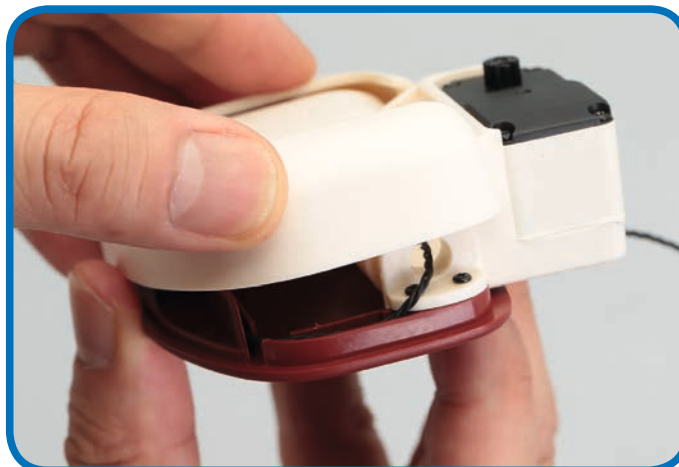
If the test does not run as described, read the troubleshooting section in this stage to see how to address the issue.

FITTING THE LEFT FOOT COVER



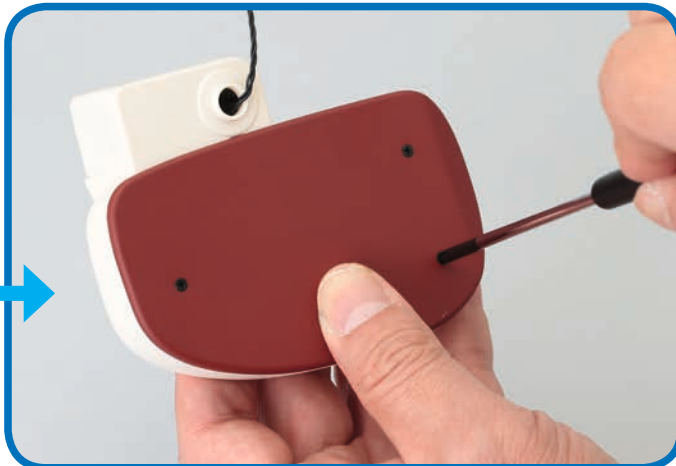
- 29** Position the left foot cover above the existing foot assembly, as shown.

- 30** Lower this to the foot assembly, so that the moulded mounting lug lines up to the arrowed hole in the sole.



- 31** At the other end of the assembly, ensure that the servo cable seen above is still neatly tucked into its groove and will not get caught.

- 32** Then, when all is in place, press the cover over the foot to fit snugly on the sole.



- 33** Holding the parts together, turn the foot over and tighten three M2 x 7mm countersunk screws into the circled holes in the sole to secure the parts.



- 34** Lay the foot out flat on your work surface, then lower the left ankle panel onto the open section of the ankle frame containing the servo cable.

- 35** Use two M2 x 6mm countersunk screws to secure the panel.



- 36** To complete this stage, lift the ankle frame to its neutral position, perpendicular to the sole.

NOTE: As with other stages in which you connected two or more servos, it is vital that you do not attach the servos linked in this stage to the test board! Doing so will wipe their ID numbers, meaning you will have to go back and open each servo to reset it.

Assembled left foot

At the completion of this stage, Robi's left foot has taken shape, with all the foot covers in place and the foot and ankle servos linked, ready for the next phase of the build.



TIP!



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

TROUBLESHOOTING

As you may have already found with Robi's left and right arms, there is a chance that the foot and ankle subassembly will not work as intended the first time it is switched on. If the movement test carried out in this stage does not go as described, the most likely cause will be an error with one of the subassembly's connectors or servos. Read this troubleshooting guide carefully to see how to assess and correct any potential issues.

Along with the Phillips screwdriver supplied with Stage 2, you will need the neck stand and test board from Stage 8 to deal with any problems that may occur with the left foot and ankle subassembly.

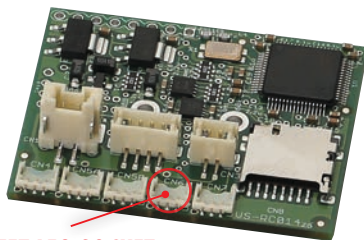
Neck stand and test board
(Stage 8)



IF NONE OF THE SERVOS MOVE

Before dismantling the foot or ankle to check the individual servos, check the other electrical components used in the test. Then, if that doesn't solve the problem, proceed to see how to safely and efficiently dismantle the arm.

CHECK THAT THE FOOT IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



LEFT LEG SOCKET

The socket that is second from the micro SD card slot on the CPU board should hold the servo cable leading to the left foot.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the left foot's servo cable connector is sitting securely within its socket, removing and replacing it carefully if need be.

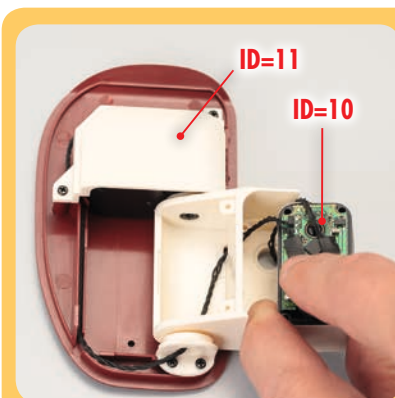
CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged by following the steps explained in Stage 41.

IF NEITHER SERVO MOVES

If neither servo moves, there is a chance that either there is a faulty connection between them, or that their ID numbers have been set incorrectly.

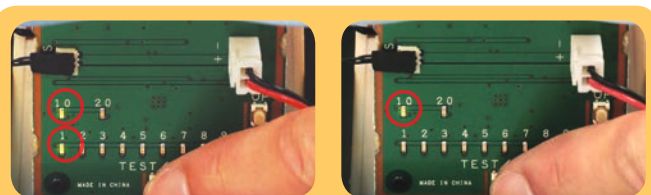


DISASSEMBLY

Begin by following the assembly steps 20-24 of this stage in reverse, and check each connection carefully. If this does not solve the issue, you will need to dismantle the foot and ankle in full and test/re-set the faulty servo(s) as necessary.

WHEN ONLY ONE SERVO MOVES

If one servo works but the other does not, it is probable that the fault is down to an ID numbering error.



DISASSEMBLY

Once the foot and ankle subassembly is fully dismantled, use the test board to check/reset the servos' ID numbers. The foot servo should be ID=11, and instructions on how to test and set it to this number are found in Stage 35. The ankle servo is ID=10, and instructions on testing and resetting this came in Stage 44.

STAGE 46: ASSEMBLE ROBI'S LEFT LOWER LEG, THEN TEST HIS RIGHT

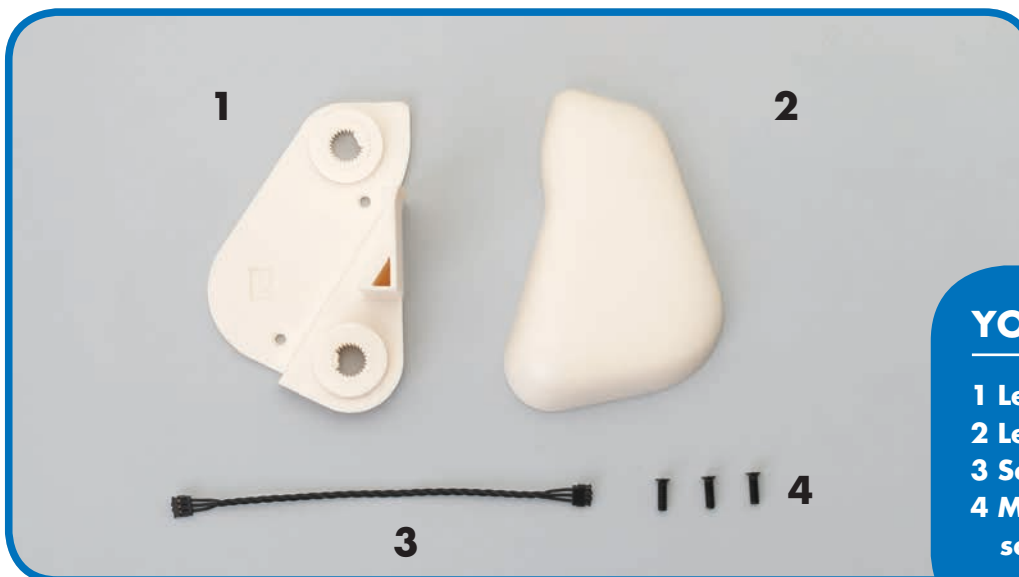
In this stage, you will begin to assemble Robi's left lower leg, before testing the movement of the already assembled right lower leg using the CPU board.

Now that you have fitted and tested the foot and ankle servos, completing the assembly of Robi's left foot, it is time to add the parts that will connect this foot to the rest of his leg. To do this, you fit the lower leg frame that will join Robi's

ankle servo to his knee servo, before adding the outer lower leg cover. Remember to take care when fitting the servo shaft into the D-cut hole in the lower leg frame, and follow the test instructions carefully.



PARTS TO BE ASSEMBLED



YOUR PARTS

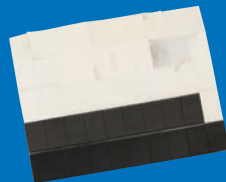
- 1 Left outer lower leg cover
- 2 Left lower leg frame
- 3 Servo cable (70mm)
- 4 M2 x 6mm countersunk screws x 3 (one is a spare)

SAVED PARTS

You will be using parts from previous stages, so be sure to have these to hand before beginning.



*Left foot assembly
(Stage 45)*



*Protective pads
(Stage 3)*



*Battery
(Stage 38)*

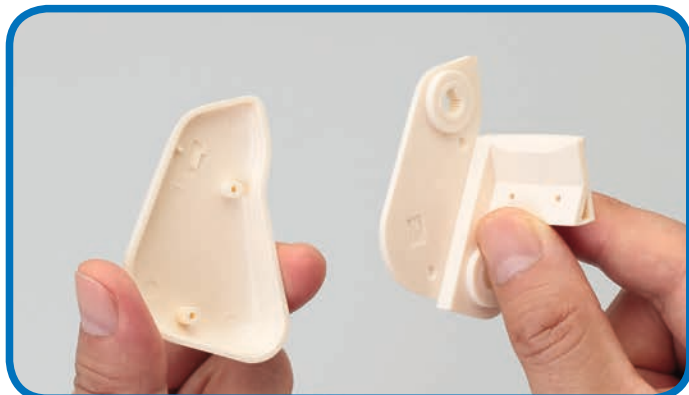


*CPU board
and support
(Stage 44)*



*Right leg
(Stage 32)*

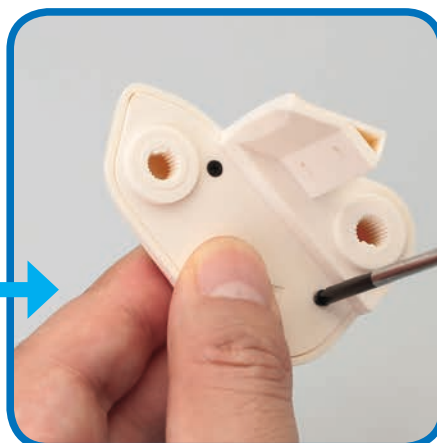
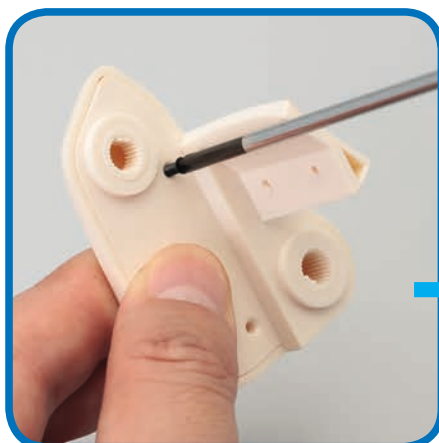
FITTING THE LOWER LEG FRAME



1 Line up the lower leg frame with its outer cover, exactly following the orientation shown in the photo.

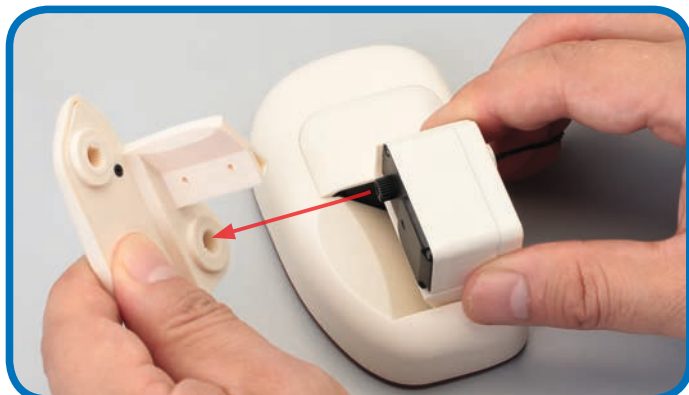


2 Press the parts together, as shown. They should fit neatly into one another.



3 Use two M2 x 6mm countersunk screws to secure the parts.

Note the D-cut holes seen on the inside of the lower leg frame – the lower hole will be the one into which you will fit the corresponding servo shaft in this stage.



4 Take the left foot assembly (Stage 45), then line up the D-cut hole in the lower leg frame with the shaft of the ankle servo. The front panel, Robi's 'shin', will stretch across the forward-facing side of the servo.

TIP!

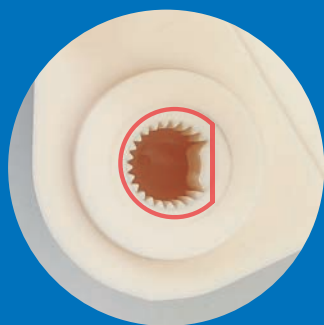
Robi's ankle will need to be fully upright and perpendicular to his foot for you to fit the lower leg frame easily. Make sure it is in this position before you continue with this stage's build.



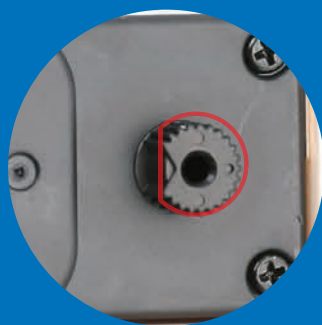
THE D-CUT



It is very important that you take care when working with the servos. Each joint, such as the lower leg frame you will be fitting in this stage, has a precisely moulded D-shaped hole, designed to fit snugly onto the D-shaped servo shaft to ensure that the servo is set correctly. When these parts are aligned as they should be, it will be easy to fit them together. If it is in any way difficult to join the parts, separate them, check that you have both of them lined up accurately, and try again. Never force the parts together, as this could damage both of them.

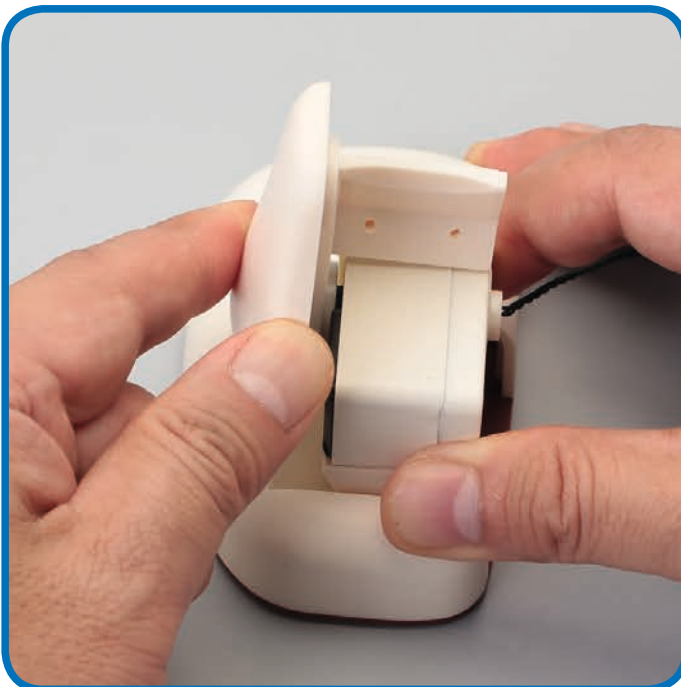


The D-shaped hole in the lower leg frame, supplied with this stage.



The servo's shaft is also D-shaped, and is designed to fit snugly into the hole in the frame.

Never force the parts together, because there is a high risk that you will damage the small plastic teeth inside the hole, as seen below.



- 5** Making sure that the D-shaped shaft and hole are set perfectly in line with one another, fit the lower leg frame onto the ankle servo's shaft.

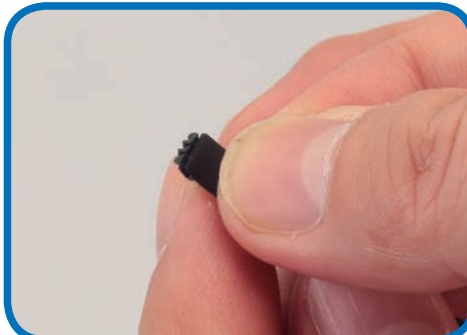
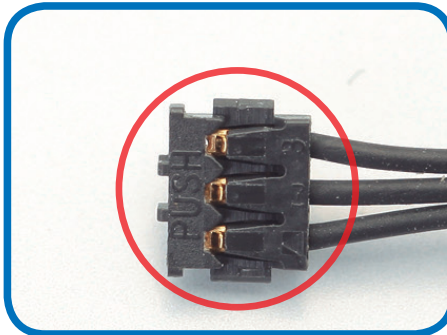
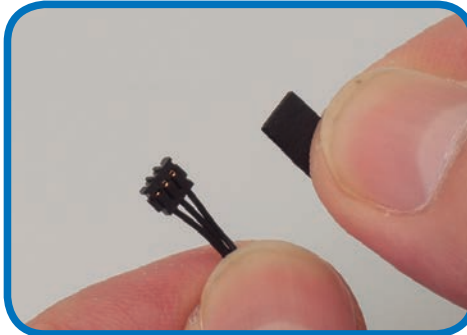
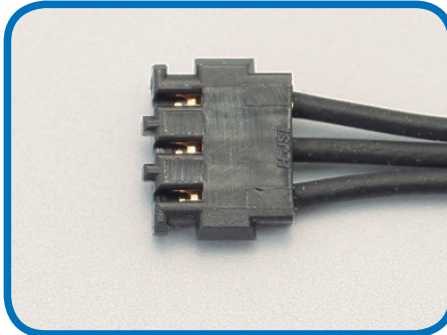


- 6** Press the parts together firmly but carefully. They will slip into place easily if correctly set – never force them.

SEALING THE SERVO CABLE

Assembled left foot and lower leg

Robi's left foot has taken shape, matching the already assembled right foot, with all the covers in place and the foot and ankle servos linked, ready for the next phase of the build.



7 As you have done previously, identify the side of this stage's servo cable connector that has PUSH marked on it (circled).

8 Apply a protective pad to the connector on this side. Repeat for the other end of the cable.

TIP!

If you want to wait to fit the pads until the cable is used in the build, you can do so.

RETESTING THE SERVOS



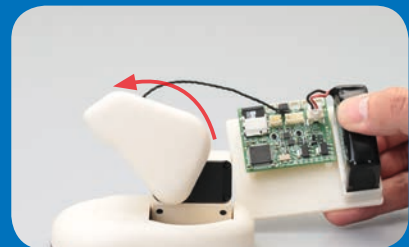
You tested the movement of the foot and ankle servos in Stage 45, but at that stage, the ankle servo wasn't attached to the lower leg frame fitted in this stage. To get an idea of how the parts will work together now that they are joined, you may like to perform the test again. If anything is amiss and you need to dismantle the subassembly, refer to the relevant steps of Stage 45; if you find you need to reset the servos' ID numbers, Stage 35 (for the foot servo, ID=11) and Stage 44 (for the ankle servo, ID=10) will help you.



Connect the ankle servo's cable to its designated socket on the CPU board (see diagram in Stage 43). Then connect the battery.



The test will start immediately, beginning by returning the servos to their neutral positions, then tilting the foot servo from side to side.



Now you will see the ankle servo in action, as it will bend the lower leg frame forwards then back, completing the test.

TESTING THE RIGHT LEG

ATTENTION!

Robi's right leg will start moving as soon as you begin the test, so make sure you perform this test in a safe place where the leg will not knock anything over.

WARNING!

When you connect the battery to the CPU board, disconnect it immediately if you see any smoke or vapour rising from the board!



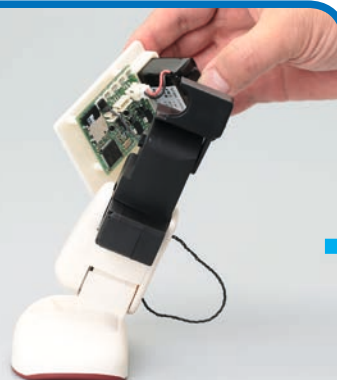
9

Fetch the right leg from Stage 32, and insert its servo cable into the right leg socket, closest to the micro SD card slot.

10

Set the right leg flat on your work surface, then connect the battery – the test will start as soon as the power is connected, so be prepared and observe the following steps.

STARTING POSITION



STARTING POSITION

11

As soon as the battery is connected, the test will begin. After the servos have all reverted to their starting positions, the first servo to move will be the foot servo. This will tilt the leg inwards at the angle shown above, then return it to the starting position once more. View this process from the front for the best view of the parts' movements.

STARTING POSITION



STARTING POSITION

12

When the foot servo has moved the leg from side to side, those in the ankle, knee and thigh will begin. These will be best viewed from the side. As with earlier tests, simply remove the battery connector from the CPU board to stop the test.

If the test does not run as described, read the troubleshooting section in this stage to see how to address the issue.

TROUBLESHOOTING

As you may have already found with Robi's other limbs, there is a chance that the right leg will not work as intended the first time it is switched on. If the movement test carried out in this stage does not go as described, the most likely cause will be an error with one of the subassembly's servos. Read this troubleshooting guide carefully to see how to assess and correct any potential issues.

Along with the Phillips screwdriver supplied with Stage 2, you will need the following:



Servo removal tool (Stage 41)

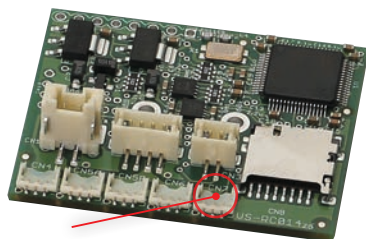


Neck stand (Stage 8)

IF NONE OF THE SERVOS MOVE

Before dismantling the leg to check the individual servos, check the other electrical components used in the test. Then, if that doesn't solve the problem, here's how to dismantle the subassembly safely and efficiently.

CHECK THAT THE LEG IS CONNECTED TO THE CORRECT SOCKET ON THE CPU



RIGHT LEG SOCKET

The socket next to the micro SD card slot on the CPU board will hold the right leg's servo cable.

CHECK THAT THE SERVO CABLE IS SECURELY CONNECTED TO THE CPU



Ensure that the right leg's servo cable connector is sitting securely within its socket. Remove and replace it carefully if need be.

CHECK THAT THE BATTERY IS CHARGED



The test will only run if sufficient power is received from the battery, so make sure the battery is fully charged by following the steps explained in Stage 41.

IF ONLY THE FOOT SERVO MOVES

If the foot servo moves but the others between it and the hip do not, there is most likely a problem with those servos' ID numbers.

DISASSEMBLY

The leg can be dismantled from the foot, knee or thigh – so you only need begin dismantling the leg at the relevant point. Remember to check the connections as you take the servos apart, as a poor connection could also be the problem.

- **The ankle servo does not move**
Dismantle the foot subassembly and check/reset the ankle servo's ID number
- **The knee servo does not move**
Dismantle the leg's knee and lower leg frames and check/reset the knee servo alone
- **The thigh servo does not move**
Dismantle the leg's thigh and hip frames and check/reset the thigh servo alone

IF THE FOOT SERVO DOES NOT MOVE

If the foot servo does not move, it could be a fault with any of the leg's servos.

DISASSEMBLY

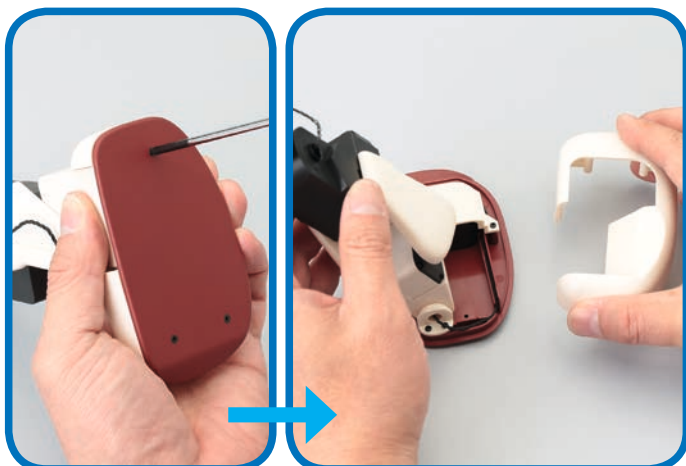
In this case, every faulty servo needs to be checked and potentially reset. Again, remember to check the connections as you take the servos apart, in case one of these is the problem – it will save you time.

- **The ankle servo does not move**
Dismantle the foot subassembly and check/reset the ankle servo's ID number
- **Neither the ankle nor the foot servos move**
Dismantle the leg from the foot to the knee and check/reset the foot and ankle servos' ID numbers
- **None of the servos move**
Dismantle the leg and check the connections between, and the ID numbers of, every leg servo

DISMANTLING THE RIGHT LEG

The steps on the following pages show how to dismantle the right leg – but rather than dismantling the leg in its entirety all at once, as you did for the arms in previous stages, it is possible to remove the leg's servos as and when required. So, if you are experiencing problems with the foot servo alone, follow **Steps 1-11**. If the ankle servo is also faulty, continue on to **Step 15**.

However, if the foot and ankle servos are OK, but the knee servo needs attention, you can start at **Step 16** and work through to **Step 23**. Lastly, if it is the uppermost thigh servo you need to check/reset, you only need to follow **Steps 24-27**. Instructions on how to reassemble each section of leg – including the setting of the servos – are found in the original stages, which are listed at the end of each disassembly.



- 1** Remove the three screws in the base of the sole to open the foot. **Keep the screws safely, making a note of where they came from.**



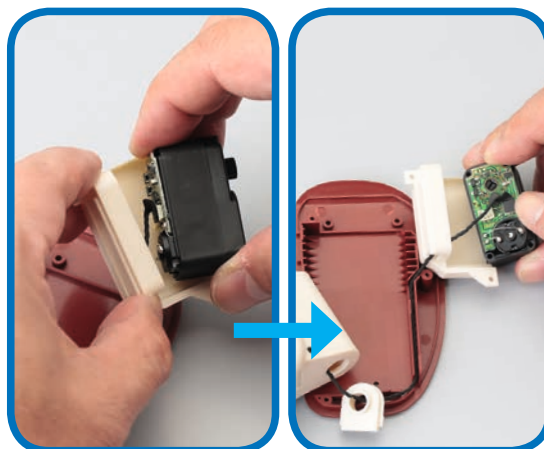
- 2** Removing the screws at the front and rear of the assembly will free the foot and ankle servos. Again, retain all the screws.



- 3** Lift the foot servo away from the sole.



- 4** Remove all four screws to unfasten the servo from its removable cover.



- 5** Gently remove the servo from its cover and check that its servo cable's connector is sitting securely within its socket.

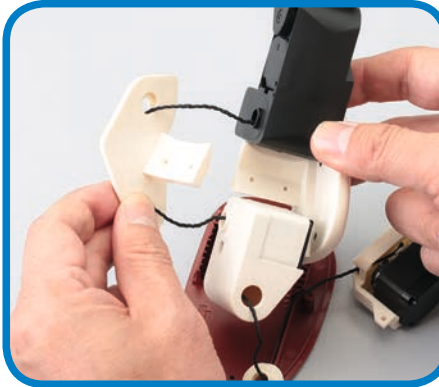
TIP!



When reinserting the servo cables' connectors as you perform the tests in this stage, always remember to fit them carefully and properly. They will click into place neatly when aligned correctly – never force them! View the servos from the side to check that the connectors are securely fastened and are flush with the servos' electronic boards.



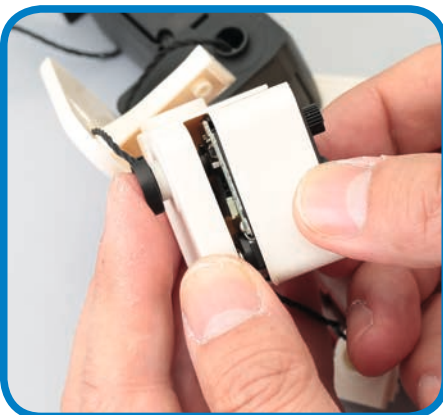
- 6** Remove the two circled screws in the rear of the lower leg parts to free the ankle servo. Again, keep the screws safely.



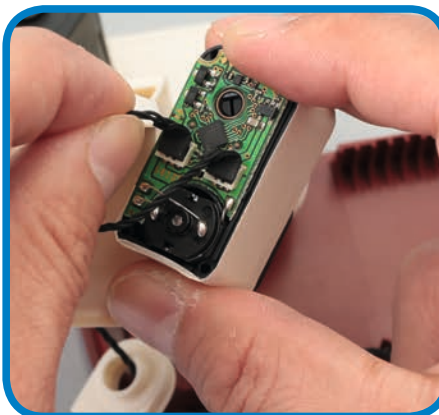
- 7** Carefully remove the lower leg inside panel. Do not tug on it, because the servo cable running through it could tear.



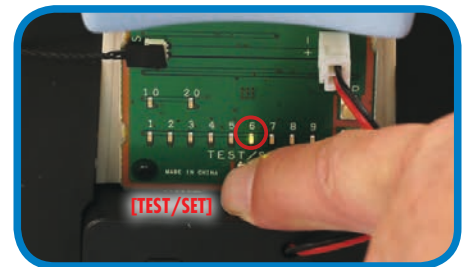
- 8** Carefully remove the ankle servo and take out the four screws to unfasten its cover.



- 9** Use the servo removal tool to open the servo.



- 10** Check the connections – if one or other is faulty, fixing it may resolve the problem. If both are sound, proceed by removing the cable leading to the foot servo.



- 11** Now you can connect the cable leading from the foot servo to the test board and check its ID number, which should be ID=6. Stage 23 gives instructions on how to set/reset this number if anything is amiss.

If it was only the foot servo that was faulty, you can now reassemble the right leg.



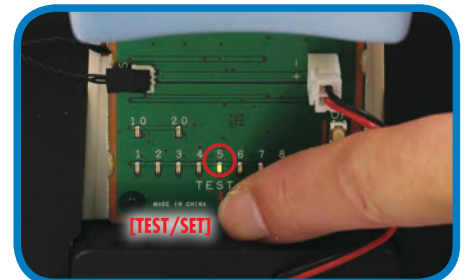
- 12** If necessary, proceed by loosening the four screws holding the knee servo in place to free it.



- 13** Carefully lift the knee servo away from its frame.



- 14** Check the connectors – if both of them are sound, remove the one that leads to the ankle servo.



- 15** You can now connect the ankle servo to the test board and check its ID number, which should be ID=5. Stage 26 lists instructions on how to set/reset this number if anything is amiss.

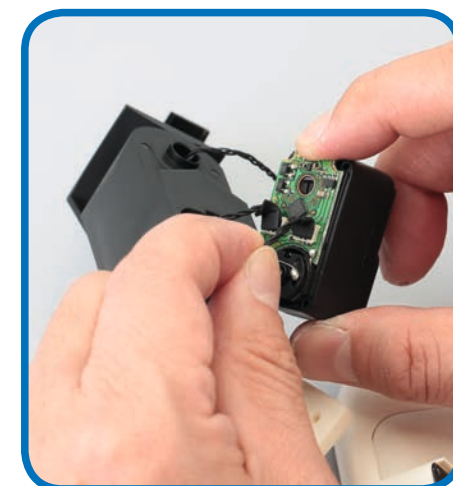
If it was only the ankle servo (or the ankle and foot servos) that was faulty, you can now proceed to reassemble the right leg.

To reassemble the right leg at this point, refer back to Stages 23-30.



16 If you need to dismantle the leg at the knee (if it is only the knee servo that needs attention), start by removing the two screws at the rear of the lower leg. Carefully separate the lower leg frame and panel to free the knee servo. Again, keep the screws, making a note of where they came from.

17 Remove the four screws securing the knee servo's cover. Keep the screws safely.



18 Now carefully remove the knee servo from its frame, being careful, as ever, not to tug on any of its servo cables. Check the connections.

19 If needed, proceed by removing the servo cable leading to the ankle servo.



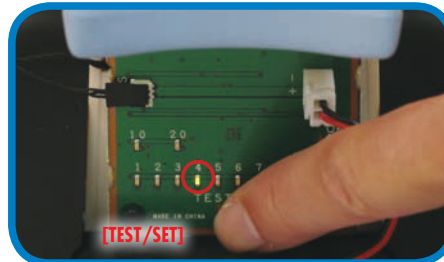
20 Next, remove the four screws holding the thigh servo in its frame. To do this, you will need to rotate the hip frame to one side then the other so that you can access the two screws at the top of the servo.



21 Use the servo removal tool to open the servo.



22 Check the connections – if one or other is faulty, fixing it may resolve the problem. If both are sound, proceed by removing the cable leading to the knee servo.

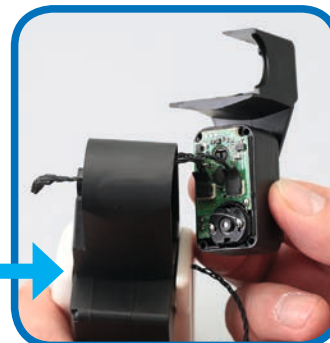


23 Now you can connect the cable leading from the knee servo to the test board and check its ID number, which should be ID=4. Stage 29 lists instructions on how to set/reset this number if anything is amiss.

To reassemble the right leg at this point, refer back to Stages 28-32.



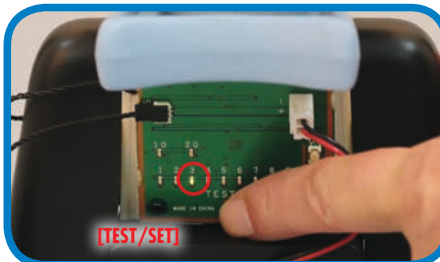
24 If you need to dismantle the leg at the thigh (if it is only the thigh servo that needs attention), remove the thigh servo from its frame by removing its four screws. First unscrew the three that are on one side of the thigh frame, then rotate the hip frame to access and remove the fourth.



25 Carefully lift the servo out of its frame using the servo removal tool.



26 Check the connections – if both are sound, proceed by removing the cable leading down to the knee servo.



27 Connect the remaining cable to the test board to check the thigh servo's ID number, which should be ID=3. Stage 31 lists instructions on how to set/reset this number if anything is amiss.

TIP!



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

To reassemble the right leg at this point, refer back to Stages 31-32.

COMING IN PACK 13

Your next **FOUR** complete Stages, as Robi starts to come to life!



Stage 47

Prepare Robi's left knee servo

PARTS PROVIDED

The servo that will move Robi's left knee.



Stage 48

Continue building Robi's left leg

PARTS PROVIDED

Robi's left knee frame and inner lower leg panel.



Stage 49

Prepare Robi's left thigh servo

PARTS PROVIDED

The servo that will move Robi's upper leg.



Stage 50

Prepare to join Robi's legs for the first time

PARTS PROVIDED

Outer and inner hip frames and a servo cable.